

Bias and Sensitivity under Ambiguity[†]

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This paper characterizes the effects of ambiguity aversion under dispersed information. The equilibrium outcome is observationally equivalent to a Bayesian forecast of the fundamental with increased sensitivity to signals and a pessimistic bias. This equivalence result takes a simple form that accommodates dynamic information and strategic interactions. Applying the result, we show that ambiguity aversion helps rationalize the joint empirical pattern between the bias and persistence of inflation forecasts conditional on household income. In a policy game à la Barro and Gordon (1983) with ambiguity-averse agents, the policy rule features higher average inflation and increased responsiveness to fundamentals. (JEL D81, D83, E31, E37, E71)

Workhorse macroeconomic models often assume that agents have perfect knowledge about the underlying data-generating process of the economy. However, households and firms often face model uncertainty or ambiguity¹ when making economic decisions. For example, after the pandemic, inflation has surged, and average inflation targeting entered the policy discussion. It is less clear whether inflation will fluctuate around 2 percent or 4 percent in the near future, and each of these scenarios could be interpreted as a different model economy. Moreover, when economic decisions are interdependent, coordination among market participants relies on their perceptions of how others would react to such uncertainty. In this environment, what are the macroeconomic effects of ambiguity? Can ambiguity help explain the deviations from rational expectations observed in survey data? How does the presence of ambiguity aversion in the private sector affect optimal policy design?

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¹According to Marinacci (2015), ambiguity refers to subjective uncertainty over probabilities due to a lack of ex ante information to determine a specific model for the economy in the course of decision-making.

This paper makes two contributions. Theoretically, we show that despite the complexities associated with dispersed information, coordination motives, and ambiguity aversion, the equilibrium strategy is equivalent to a single-agent forecasting problem with two modifications: an amplified responsiveness to signals and a permanent pessimistic bias. This equivalence result applies to general information processes and takes a simple form, enabling the use of standard numerical algorithms to compute equilibrium strategies. On the applied side, we document that the bias in inflation forecasts decreases with household income, while the persistence of forecast errors increases with household income. This observed joint distribution can be naturally accounted for in a model where agents exhibit ambiguity aversion toward inflation shocks. In a policy game à la Barro and Gordon (1983), we show that policymakers respond to this behavioral pattern by increasing both their responsiveness to shocks and the unconditional level at which they set the inflation rate.

Framework.—We consider an abstract Gaussian-quadratic economy. Agents' payoffs depend on an exogenous fundamental, their own actions, and the average actions of others, following Angeletos and Pavan (2007). The generic quadratic utility function can be regarded as the reduced-form representation of a microfounded economy that allows for general equilibrium (GE) effects. In addition, we impose minimal restrictions on the information structure to accommodate persistent learning and dispersed information. The main departure from the existing literature is that the fundamental is not only stochastic but also ambiguous, in the sense that agents do not have perfect knowledge about its objective probability distribution. In the baseline specification, agents are ambiguity averse, with preferences represented by the smooth model of ambiguity proposed by Klibanoff, Marinacci, and Mukerji (2005). We adopt the smooth rule when updating preferences (Hanany and Klibanoff 2009), ensuring dynamic consistency and optimal ex ante equilibrium strategies.² We also extend the analysis to models with robust preferences (Hansen and Sargent 2001a, b).

The interaction between ambiguity aversion and dispersed information makes the equilibrium difficult to characterize. It has been widely recognized that ambiguity-averse agents behave as if their beliefs about the fundamental are distorted (Ilut and Schneider 2014; Bhandari, Borovička, and Ho 2024) and that these distortions are affected by the equilibrium strategies. This results in a fixed-point problem between the strategies and beliefs of all agents in equilibrium. On top of that, the presence of imperfect coordination and persistent information leads to the infinite regress problem (Townsend 1983), with higher-order beliefs potentially requiring an infinite-dimensional state space.

Equivalence Results.—Conceptually, the presence of ambiguity amounts to a more diffuse prior, and ambiguity aversion makes the diffusion loom larger. As a result, agents divert attention from their priors to their signals, leading to a higher sensitivity to signals. Meanwhile, in the face of model uncertainty, ambiguity-averse agents are more concerned about models that yield lower payoffs and tend to place

²This approach is equivalent to characterizing the sequential equilibrium with ambiguity, an equilibrium refinement proposed in Hanany, Klibanoff, and Mukerji (2020).

greater emphasis on adverse probability distributions. This consideration leads agents to behave as if they are pessimistically biased when making forecasts.

Our equivalence result formalizes this intuition and circumvents the aforementioned technical complexities. The equilibrium strategy is ultimately equivalent to that of a modified single-agent Bayesian forecasting about the fundamental. We first introduce a (w, α) -modified signal process. In this auxiliary forecasting problem, the precision of idiosyncratic shocks is discounted by the degree of strategic complementarity, α , which captures the idea that the coordination motive reduces the importance of private information in inferring the aggregate outcome (Huo and Pedroni 2020). In addition, the prior variance of the fundamental shock is amplified by w , an endogenous object that summarizes the extent to which the prior becomes more diffuse due to ambiguity aversion.

We establish that the equilibrium strategy coincides with the Bayesian forecasting rule using the (w, α) -modified signal process, with two notable adjustments governed by the endogenous variable w . First, there is an additional uniform overreaction to all signals. This overreaction distinguishes the equilibrium strategy from a Bayesian forecasting problem, resembling the departure from rationality implied by diagnostic expectations (Bordalo et al. 2020). Second, there is an additional bias that is independent of the signal realization, further differentiating the equilibrium allocation from its rational-expectations counterpart. To close the loop, we provide the condition that w needs to satisfy, which involves only unconditional moments about endogenous aggregate outcomes.

This result provides a concise summary of the effects of ambiguity aversion in a general equilibrium setting. The characterization enables the derivation of general comparative static results. For example, we show that an increase in the degree of strategic complementarity leads to greater bias. To understand the underlying intuition, one needs to invoke the fact that, with ambiguity aversion, equilibrium outcomes depend on *subjective* higher-order beliefs. With rational expectations, beliefs of higher order rely more on the common prior (Morris and Shin 2002). With ambiguity aversion, the bias in others' beliefs is embedded in the common prior, which then accumulates as the order increases. A higher degree of strategic complementarity increases the relative weight of higher-order beliefs, thereby amplifying the bias.

Computationally, this result provides a tractable algorithm for computing the equilibrium strategies. In the absence of ambiguity, dynamic higher-order expectations require, in principle, the entire history of signals as state variables. However, a finite-state representation is possible when signals follow ARMA(p, q) processes (Woodford 2003; Angeletos and La'O 2010; Huo and Pedroni 2020). Our results imply that a similar finite-state equilibrium representation is still possible even with ambiguity-averse agents. The original infinite-dimensional problem collapses to a one-dimensional problem to determine the endogenous amplifier of the prior variance, w . To solve the Bayesian forecasting problem with the (w, α) -modified signal process, the standard Kalman filter can be applied.

Survey Evidence on Inflation Expectations.—Our equivalence result helps explain the patterns observed in survey data regarding the joint behavior of bias and persistence in inflation forecasts. Using the Michigan Survey of Consumers (MSC)

and the Survey of Consumer Expectations (SCE), we document that the bias and the persistence of inflation forecast errors jointly vary with household income levels. Specifically, higher-income households tend to exhibit lower forecast bias but more persistent forecast errors. This joint behavior of bias and persistence is at odds with the predictions of rational expectations models. Rationality would imply an average bias of zero, and the higher persistence of forecast errors of the rich would require that they have less precise information.

We demonstrate that the observed patterns naturally arise from a microfounded optimal consumption problem in which households face ambiguity about the inflation process. With rigid nominal incomes, higher inflation erodes households' real purchasing power. Accordingly, ambiguity-averse households assign more weight to high-inflation models, resulting in an upward bias in inflation forecasts. At the same time, ambiguity aversion induces increased sensitivity to signals or a reduction in the weight put on the prior mean of inflation, which in turn reduces the persistence of forecast errors.

Importantly, households with lower nominal incomes are more exposed to variations in inflation and are more sensitive to ambiguous inflation shocks. In the cross section of households, it is as if higher-income households were less ambiguity averse. Consequently, the inflation forecasts of the income rich feature relatively lower levels of bias, lower sensitivity to signals, and higher persistence of forecast errors. With this rather parsimonious structure, our model matches the documented cross-sectional patterns reasonably well when calibrated to the data. Further, the predictions of the model are also broadly consistent with existing survey evidence on expectations, including underreaction at the consensus level (Coibion and Gorodnichenko 2015), overreaction at the individual level (Bordalo et al. 2020), and delayed overshooting (Angeletos, Huo, and Sastry 2021).

Optimal Policy with Ambiguity Aversion.—How should policy respond to the observed deviations from rational expectations? To shed light on this question, we explore a policy game à la Barro and Gordon (1983). Specifically, a policymaker attempts to strike a balance between minimizing the unemployment rate and minimizing deviations from the inflation target, subject to the Phillips curve. The optimal inflation policy is then a weighted average of the exogenous random inflation target and the average inflation expectation in the private sector. Departing from Barro and Gordon (1983), we allow agents to receive dispersed information and face ambiguity about the inflation target. This ambiguity may arise from factors such as ambiguous policy communication or a loose policy objective like average inflation targeting.

The optimal policy rule is an affine function of the exogenous inflation target. The slope represents the responsiveness to changes in the target, while the intercept determines the long-run average inflation rate. In the presence of imperfect information but no ambiguity, the slope is dampened due to the underreaction of the consensus forecasts, but the intercept coincides with the mean of the inflation target. Introducing ambiguity results in additional sensitivity to information and a permanent positive bias, so a steeper slope and a lifted intercept in the policy rule. The former brings the inflation dynamics closer to the first best, while the latter pushes the policy away from it. We show that it is always beneficial to have some ambiguity

about the inflation target, as the initial benefits from increased sensitivity outweigh the initial costs associated with the higher bias. However, for high-enough ambiguity, this relationship flips, and additional ambiguity reduces welfare. It follows that an intermediate level of ambiguity is desirable.

Robust Preferences.—Although our analysis focuses on the smooth model of ambiguity, the main insights on bias and sensitivity extend to models with robust preferences. Despite these being significantly different approaches to ambiguity, we find that under certain conditions, for a robust preferences model, a corresponding smooth model of ambiguity exists such that the equilibrium strategies in both models are identical up to a constant. There are subtle differences in determining w for the auxiliary (w, α) -modified signal process,³ but the quantitative predictions of the two models in our applications are notably similar.

Related Literature.—This paper contributes to the literature exploring the implications of ambiguity and ambiguity aversion in macroeconomic models. There are three prominent representations of ambiguity-averse preferences in the literature: (i) the multiple priors preference axiomatized by Gilboa and Schmeidler (1989); (ii) the robust preferences proposed by Hansen and Sargent (2001a,b); and (iii) the smooth model of ambiguity axiomatized by Klibanoff, Marinacci, and Mukerji (2005). Each of these representations has been extensively used in macroeconomic applications. For example, in the context of business cycle models, Ilut and Schneider (2014); Bianchi, Ilut, and Schneider (2017); and Ilut and Saijo (2021) employ the multiple priors preference approach.⁴ Luo and Young (2010); Bidder and Smith (2012); and Bhandari, Borovička, and Ho (2024) use robust preferences. Backus, Ferriere, and Zin (2015); Altug et al. (2020); and Pei (2024) apply the smooth model of ambiguity. In the asset pricing literature, Epstein and Wang (1994); Chen and Epstein (2002); and Miao (2009) use the multiple priors preference approach. Hansen, Sargent, and Tallarini (1999) and Anderson, Hansen, and Sargent (2003) utilize robust preferences. Ju and Miao (2012); Collard et al. (2018); and Gallant, Jahan-Parvar, and Liu (2018) employ the smooth model of ambiguity. Additionally, Michelacci and Paciello (2019) study the effects of monetary policy announcements under multiple priors preferences.

Most of the aforementioned works assume representative agents and abstract from incomplete information. We analyze the effects of ambiguity and ambiguity aversion within a flexible environment that accommodates not only GE considerations but also incomplete information and persistent learning. In this regard, our findings complement the literature on games with incomplete information (Morris and Shin 2002; Woodford 2003; Angeletos and Pavan 2007; Angeletos and La'O 2010), extending the analysis beyond the rational expectations benchmark. Our theoretical results also establish a link between the equilibrium outcomes in the smooth model of ambiguity and the robust preferences model. In this vein, Cerreia-Vioglio, Corrao,

³ With robust preferences, the condition that w needs to satisfy involves conditional moments about individual variables, whereas the condition in the smooth model involves only unconditional moments of aggregate variables.

⁴ We refer to Ilut and Schneider (2023) for a more comprehensive review of recent development in the applications of multiple priors preferences in macroeconomics.

and Lanzani (2024) characterize the effects of ambiguity in the high-coordination limit under general preferences, while our results apply for any level of coordination when restricting attention to more stylized preferences.

Our paper is also related to an extensive body of literature examining systematic biases in agents' expectations using survey data. Elliott, Komunjer, and Timmermann (2008) present evidence of systematic bias in professional forecasters' expectations and suggest that an asymmetric loss function rationalizes the documented biases.⁵ Similar evidence of biased expectations has been documented by Kohlhas and Robertson (2024) and Farmer, Nakamura, and Steinsson (2024) using the Survey of Professional Forecasters. Kohlhas and Robertson (2024) show that professional forecasters' expectations are biased but more accurate than commonly used time series models, particularly in the short run. They propose a theory of cautious expectations in which agents estimate the optimal weight on observed signals using classical inference, resulting in a trade-off between bias and accuracy. Farmer, Nakamura, and Steinsson (2024) present evidence of bias in macroeconomic expectations using the Survey of Professional Forecasters (SPF) and note that the associated forecast errors are serially correlated. Within a Bayesian paradigm that retains rationality, the authors demonstrate that slow learning over the long-run trend with a unit root can rationalize the documented bias and persistence in forecast errors. In a similar context, Andolfatto, Hendry, and Moran (2008) argue that the bias in inflation expectations can arise from small sample problems. In a recent study, using machine learning algorithms, Bianchi, Ludvigson, and Ma (2022) also document substantial bias in macroeconomic expectations of professional forecasters and reveal its cyclical properties. Relatedly, Azeredo da Silveira, Sung, and Woodford (2024) and Sung (2024) show that imprecise memory can lead to biased forecasts and have rich implications for sensitivity depending on forecasting horizons.⁶

The evidence of biased expectations extends beyond professional forecasters. Using the MSC, Bhandari, Borovička, and Ho (2024) document that households' inflation and unemployment rate forecasts feature pessimistic biases, which are countercyclical and co-move positively along the business cycle. Rozsypal and Schlafmann (2023) document a systematic bias in individual-level income expectations that varies with income levels.⁷ They argue that an overpersistence bias rationalizes this evidence, which is broadly consistent with the approach proposed by Molavi (2023). Using survey evidence in the United Kingdom, Michelacci and Paciello (2024) study how the pessimistic bias in inflation forecasts is related to households' wishes about inflation and nominal interest rates.

We contribute to this literature by presenting evidence on the joint behavior of bias and persistence of forecast errors across the income distribution, which cannot be easily rationalized by existing theories of expectation formation. Our theory

⁵ Pope and Schweitzer (2011) demonstrate that bias originating from loss aversion can persist even in high-stakes contexts using data on the performance of professional golfers in the PGA TOUR.

⁶ The notion of bias in Azeredo da Silveira, Sung, and Woodford (2024) and Sung (2024) is more related to the comparison between sensitivities under rational expectations and bounded rationality, while our notion of bias refers to the deviations of the unconditional mean of forecast errors from the rational benchmark.

⁷ See Dominitz and Manski (1997); Dominitz (1998); Das and van Soest (1999); and Massenot and Pettinicchi (2019) for more related studies on the pessimistic bias of expectations on individual income.

provides a joint characterization of bias and sensitivity, directly addressing these empirical patterns.

Finally, our paper is related to the large literature on optimal policy under incomplete information (Adam 2007; Lorenzoni 2010; Paciello and Wiederholt 2014; Amador and Weill 2010; Angeletos and La'o 2020; Angeletos and Sastry 2021). This line of work has mainly focused on the responsiveness and cyclicalities of the policy instrument when agents are subject to informational frictions, while the mistakes made by the private sector are mostly temporary. Relative to the existing literature, the additional bias and sensitivity in our environment introduce a trade-off between short-run responsiveness and long-run bias in policy design, and we illustrate the different welfare implications with alternative microfoundations for the presence of biases.

I. An Illustrative Example

In this section, we discuss the interaction between ambiguity and imperfect information in a single-agent environment. We show how ambiguity aversion increases sensitivity to signals and leads to biased forecasts, relative to the Bayesian benchmark. We then extend these insights to a more general setting in the next section, where we introduce recurrent shocks and general equilibrium considerations.

A. A Pure Forecasting Problem

Consider an inference problem about some exogenous economic fundamental. Assume the fundamental ξ is drawn from a Gaussian distribution with mean $\bar{\mu}$ and variance σ_ξ^2 ,

$$\xi \sim \mathcal{N}(\bar{\mu}, \sigma_\xi^2).$$

Agent i does not observe the fundamental perfectly but receives a private noisy signal x_i about it:

$$x_i = \xi + \epsilon_i, \text{ with } \epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2).$$

In the absence of ambiguity, the distributions of underlying shocks are common knowledge. Agents perceive that there is a single model, or a single probability distribution, that describes the stochasticity of random variables. Uncertainty arises from the realization of shocks within a model, which is referred to as risk.

Bayesian Expectations.—To begin, we revisit the standard Bayesian benchmark. In this case, agents simply want to minimize the mean-squared error (MSE) of their forecast. With Gaussian shocks, we can focus on linear strategies,

$$g(x_i) = sx_i + b,$$

where the s represents the sensitivity to signals and b is some constant. Given a particular strategy characterized by the pair (s, b) , the MSE can be expressed as the sum of two terms

$$(1) \quad E\left[(g(x_i) - \xi)^2\right] = \mathcal{R}(s) + [b - (1 - s)\bar{\mu}]^2,$$

$$\text{with } \mathcal{R}(s) = s^2\sigma_\epsilon^2 + (1 - s)^2\sigma_\xi^2.$$

The first term, $\mathcal{R}(s)$, represents the cost of within-model uncertainty (risk), a weighted sum of the variance of the fundamental and of the noise. For any sensitivity s , the second term can always be set to zero by appropriately choosing the constant b . It follows that the optimal sensitivity, s^{RE} , is obtained by minimizing the cost of risk:

$$s^{RE} = \arg \min \mathcal{R}(s) = \frac{\sigma_\xi^2}{\sigma_\xi^2 + \sigma_\epsilon^2}.$$

The optimal strategy is then given by

$$(2) \quad g(x_i) = s^{RE}x_i + (1 - s^{RE})\bar{\mu},$$

a simple weighted average between the prior mean and the signal, following Bayesian updating. The optimal sensitivity corresponds to the familiar signal-to-noise ratio.

B. Ambiguity and Ambiguity Aversion

Ambiguity.—When agents face ambiguity, they could perceive multiple plausible models that describe the economy, each corresponding to a different distribution of underlying shocks. This generalization accommodates the possibility that agents may have doubts about what is the right model of the economy. For instance, in the aftermath of the pandemic, consumers may be uncertain about whether inflation will fluctuate around an average level of 2 percent or 4 percent in the upcoming years. Similarly, during the slow recovery from the Great Recession, firms may wonder whether their sales growth will remain stagnant or rebound to its pre-recession level.

We restrict our attention to the case where agents face ambiguity about the prior mean of the aggregate fundamental ξ .⁸ Objectively, ξ is distributed according to $\mathcal{N}(\bar{\mu}, \sigma_\xi^2)$. Subjectively, however, agents' priors do not necessarily coincide with the objective distribution. They believe there can be multiple possible prior means, which themselves follow a normal distribution centered around the objective mean:

$$\xi \sim \mathcal{N}(\mu, \sigma_\xi^2), \text{ where } \mu \sim \mathcal{N}(\bar{\mu}, \sigma_\mu^2).$$

⁸ We explore ambiguity about the variance of the fundamental in online Appendix G.1. In this scenario, the equilibrium strategy exhibits higher sensitivity but zero bias. Additionally, in online Appendix G.2, we investigate the possibility that agents perceive ambiguity about the variance of the noise. In this case, we find that the equilibrium strategy exhibits less sensitivity rather than more, but still exhibits zero bias.

In this case, different models are indexed by different μ , while σ_μ^2 parameterizes the ex ante uncertainty about the models. When $\sigma_\mu^2 = 0$, we return to the Bayesian benchmark. Without loss of generality, we assume that $\bar{\mu} = 0$. The ex ante probability density function of the perceived distribution of μ satisfies

$$p(\mu) \propto \exp\left(-\frac{1}{2}\sigma_\mu^{-2}\mu^2\right).$$

Ambiguity Aversion.—We now specify the preference of agents toward ambiguity. Specifically, we are interested in the case where agents are ambiguity averse, meaning they dislike ambiguity more than risk. To this end, given a strategy $g(x_i)$, similarly to Klibanoff, Marinacci, and Mukerji (2005), we adopt the following loss function $\mathcal{L}(g)$,

$$(3) \quad \mathcal{L}(g) = \phi^{-1}\left(\int_\mu \phi\left(E^\mu\left[(g(x_i) - \xi)^2 - \chi\xi\right]\right)p(\mu)d\mu\right).$$

The integral over μ reflects the fact that agents face additional uncertainty about the prior mean. For each μ , $E^\mu[\cdot]$ denotes the mathematical expectations in an economy where the prior mean is given by μ . We assume that agents can commit to following the strategy $g(x_i)$, which is equivalent to adopting the smooth rule when updating preferences, ensuring dynamic consistency (Hanany and Klibanoff 2009).⁹

There are two additional modifications relative to the previous forecasting problem. First, the transformation $\phi(\cdot)$ introduces an additional cost associated with ambiguity about μ . When $\phi(\cdot)$ is linear, the problem reduces to

$$(4) \quad \mathcal{L}(g) = \int_\mu E^\mu\left[(g(x_i) - \xi)^2 - \chi\xi\right]p(\mu)d\mu.$$

This corresponds to the *ambiguity-neutral* case, where the uncertainty about μ is treated in the same way as the within-model uncertainty about ξ and ϵ_i . Namely, adding uncertainty about μ is equivalent to drawing a compound lottery.

The distinction between ambiguity and risk becomes meaningful when the transformation $\phi(\cdot)$ is convex. Then, agents incur additional losses when their perceived model is found to be incorrect, and agents are said to have ambiguity aversion. In what follows, we assume that $\phi(\cdot)$ takes a constant absolute ambiguity aversion (CAAA) form, that is,

$$(5) \quad \phi(x) = \frac{1}{\lambda}\exp(\lambda x),$$

where $\lambda \geq 0$ measures the degree of ambiguity aversion.

Second, the reduced-form quadratic loss function can be derived from a general decision-making problem in which the optimal strategy is equivalent to forecasting the solution under complete (or perfect) information; this is the approach we follow in Section II. Note that the payoff directly depends on the level of the fundamental, captured by the term $\chi\xi$. For example, firms benefit from higher TFP regardless of their production decisions, while consumers benefit from higher real

⁹More details are provided in the remark at the end of Section IIA.

income independent of their consumption-saving decisions. The direct dependence on the exogenous fundamental is common in economic problems but can usually be ignored, as it is inconsequential when agents are ambiguity neutral or when there is no ambiguity. In contrast, this dependence leads to biased forecast when agents are ambiguity averse.

Sensitivity and Bias.—To see how ambiguity aversion modifies agents' strategies, it is useful to decompose the loss function into the costs due to risk and ambiguity. Given a linear strategy $g(x_i) = sx_i + b$,¹⁰ this decomposition can be expressed as

$$(6) \quad \mathcal{L}(g) = \underbrace{\mathcal{R}(s)}_{\text{cost of risk}} + \underbrace{\frac{1}{\lambda} \log \int_{\mu} \exp \left\{ \lambda \left[(b - (1 - s)\mu)^2 - \chi\mu \right] \right\} p(\mu) d\mu}_{\text{cost of ambiguity}}.$$

First, ambiguity aversion leads to a higher sensitivity toward signals. Intuitively, a more diffused prior leads agents to rely more heavily on signals, and ambiguity aversion amplifies this effect. To see the forces more clearly, note that the cost of risk, $\mathcal{R}(s)$, remains the same as in equation (1). When there is no ambiguity, an agent can simply choose $s = s^{RE}$ to minimize $\mathcal{R}(s)$. When uncertainty about the prior mean is present, a trade-off arises between the cost of risk and the cost of ambiguity. At one extreme, if an agent only wants to minimize the cost of ambiguity, they would set $s = 1$ to eliminate the impact of ambiguity on forecast errors. Striking a balance between the two types of cost implies an enhanced sensitivity toward signals.

Second, ambiguity aversion introduces a bias in agents' strategies and forecasts. In equation (6), the direct effect of the exogenous fundamental, $\chi\mu$, is not symmetric around zero. Consider the case where $\chi > 0$. The losses in a model indexed with a negative prior mean are higher than the gains in a model indexed with a positive prior mean. Similar to the idea of self-insurance, an agent finds it optimal to mitigate losses in "rainy days" by assigning more weight to models with a negative μ . This affects the choice of the constant b , leading to biased forecasts. Ex ante, this incentive makes it appear as if agents have a more pessimistic view of the world.

The adjustment of sensitivity and bias in an agent's strategy is driven by their payoffs, as they place more weight on models that generate higher expected costs due to ambiguity aversion. Simultaneously, the magnitude of these costs is determined by the chosen strategy. This interdependence leads to a fixed-point problem. According to the following proposition, the optimal strategy can be understood as if agents faced no ambiguity but had a modified prior belief.

PROPOSITION 1: *The optimal linear strategy is equivalent to the Bayesian expectation with a more diffused prior belief, $\xi \sim \mathcal{N}(0, \tilde{\sigma}_{\mu}^2(s^*) + \sigma_{\xi}^2)$, and a bias*

$$g(x_i) = s^*x_i + \mathcal{B},$$

¹⁰ With ambiguity aversion but in the absence of strategic interactions, the unique optimal response is a linear function of signals. See online Appendix D for more details.

where the sensitivity s^* satisfies

$$(7) \quad s^* = \frac{\tilde{\sigma}_\mu^2(s^*) + \sigma_\xi^2}{\tilde{\sigma}_\mu^2(s^*) + \sigma_\xi^2 + \sigma_\epsilon^2}, \text{ with } \tilde{\sigma}_\mu^2(s) \equiv \frac{\sigma_\mu^2}{1 - 2\lambda\sigma_\mu^2(1 - s)^2},$$

and the bias is given by

$$(8) \quad \mathcal{B} = -\chi\lambda\sigma_\mu^2(1 - s^*).$$

Let us unpack these expressions. First of all, given the additional variance, $\tilde{\sigma}_\mu^2(s^*)$, in the prior belief, the optimal sensitivity is identical to the one under Bayesian expectations (2). The determination of the optimal sensitivity, s^* , requires solving a fixed-point problem. In the presence of ambiguity aversion, agents increase their sensitivity to reduce the ambiguity cost. At the same time, however, the degree to which they want to penalize the ambiguity cost is endogenously determined by the employed sensitivity. The system (7) succinctly summarizes these forces.

Next, even though the ex ante belief about the distribution of μ is centered around zero, agents behave as if the prior mean is biased, captured by $\mathcal{B}$. The “as if” shift of the mean is increasing in how much agents directly care about the fundamental, the degree of ambiguity aversion, and the amount of ambiguity.

The presence of ambiguity aversion predicts a joint pattern of sensitivity and bias. In the sequel, we show that this prediction is robust to different information structures and preferences and is supported by survey evidence on expectations.

II. General Equivalence Result

In this section, we present our main theoretical results. We extend the intuition from the illustrating example to an environment with more general preferences that also accommodate strategic interactions between agents. In addition, we allow for flexible persistent information structures and learning dynamics. We show that the optimal action under ambiguity aversion is akin to a pure forecasting problem of the fundamental with a modified prior belief.

A. Environment

Objective Environment.—The economy is populated by a continuum of agents indexed by i . Agents care about a common fundamental, ξ_t , which follows the stochastic process:

$$\xi_t = a(L)\eta_t, \text{ with } \eta_t \sim \mathcal{N}(0, \sigma_\eta^2),$$

where $a(L)$ is a polynomial in the lag operator L and η_t is the innovation to the aggregate fundamental.¹¹

¹¹ So far, η_t is the only aggregate shock in the economy. We extend the analysis to allow for multiple aggregate shocks in online Appendix B.3.

Agents receive dispersed information about the fundamental. The vector of signals observed by individual agent i every period is given by

$$(9) \quad x_{it} = m(L)\eta_t + n(L)\epsilon_{it}, \text{ with } \epsilon_{it} \sim \mathcal{N}(0, \Sigma),$$

where $m(L)$ and $n(L)$ are polynomial matrices in L that determine the dynamics of the signal process and ϵ_{it} is a vector of idiosyncratic noises that wash out in aggregate. Unlike in Section I, the fundamental as well as the signals can be persistent, which implies that the entire history of signals will be relevant for inference. To ensure that the random variables are stationary and signals do not contain future information, we make the following standard assumption.

ASSUMPTION 1: *All elements of $a(L)$, $m(L)$, and $n(L)$ contain only L with nonnegative powers and are square-summable.*

Ambiguity.—In the objective environment, η_t is normally distributed with mean zero. Subjectively, agents believe that η_t is drawn from a Gaussian distribution with the same volatility, σ_η^2 , but there is uncertainty about its prior mean, denoted by μ_t . The ambiguity about ξ_t is captured by the perception that

$$(10) \quad \eta_t \sim \mathcal{N}(\mu_t, \sigma_\eta^2), \text{ and } \mu_t \sim \mathcal{N}(0, \sigma_\mu^2).$$

As in Section I, the value of σ_μ^2 determines the degree of ambiguity.

Preference toward Risk.—We first specify preferences about within-model risk. As a baseline, we consider a class of economies with quadratic utility given by

$$(11) \quad u(k_{it}, K_t, \xi_t) = -\frac{1}{2}[(1 - \alpha)(k_{it} - \xi_t)^2 + \alpha(k_{it} - K_t)^2] - \chi\xi_t - \frac{1}{2}\gamma\xi_t^2,$$

where k_{it} denotes agent i 's action and K_t denotes the aggregate outcome of the economy,

$$K_t \equiv \int k_{it} di.$$

The first component of the utility function, $(1 - \alpha)(k_{it} - \xi_t)^2 + \alpha(k_{it} - K_t)^2$, captures the payoff directly associated with the individual agent's action. Agents aim to align their actions with the exogenous fundamental and the aggregate outcome. The degree of strategic complementarity, controlled by the parameter α , influences the strength of general equilibrium considerations. The second component, $\chi\xi_t + \frac{1}{2}\gamma\xi_t^2$, captures the nonstrategic impact of the fundamental on the agent's utility. This component only affects the agent's optimal strategy when there is ambiguity and agents are ambiguity averse.

The utility specification in equation (11) can be viewed as a quadratic approximation of a generic utility function, $u(k_{it}, K_t, \xi_t)$, as specified in Angeletos and Pavan (2007).¹² This specification accommodates strategic interactions among agents and a flexible

¹²The approximation is centered around the nonstochastic steady state: either the deterministic or the ambiguous steady state depending on the model environment.

dependence on fundamentals but excludes the dependence on $(K_t - \xi_t)^2$, K_t , and $K_t \xi_t$. In the language of Angeletos and Pavan (2007), specification (11) pertains to economies that are efficient under both complete and incomplete information. However, in general, the underlying economy may be inefficient, and in such cases, dependence on the aforementioned terms may arise. Nevertheless, our main observational equivalence result still applies in these cases, as discussed in online Appendix B.2.

Preference toward Ambiguity.—Agents are assumed to be ambiguity averse, with preferences represented by the smooth model of ambiguity,

$$(12) \quad \phi^{-1} \left(\int_{\mu^t} \phi \left(E^{\mu^t} [u(k_{it}, K_t, \xi_t)] \right) p(\mu^t) d\mu^t \right).$$

The expectation operator $E^{\mu^t} [u(k_{it}, k_t, \xi_t)]$ denotes the ex ante expected utility under the model indexed by $\mu^t \equiv \{\mu_t, \mu_{t-1}, \dots\}$, and $p(\mu^t)$ denotes the prior belief about μ^t derived from (10). We continue to assume the functional form:

$$\phi(x) = -\frac{1}{\lambda} \exp(-\lambda x),$$

which permits the tractability of the inference problem.

Remark on Ex Ante Strategy.—Similar to the illustrative example from Section I, each agent i chooses their preferred strategy, which is now a contingency plan denoted by $k_{it} = g(x_i^t)$, for every possible history of private signals, $x_i^t \equiv \{x_{it}, x_{it-1}, \dots\}$. We assume that all agents can commit to following their strategies, determined ex ante, when taking actions ex post. It is equivalent to assuming that the conditional preferences of agents upon receiving a history of private signals, x_i^t , are dynamically consistent.¹³ The smooth rule of updating proposed by Hanany and Klibanoff (2009) ensures dynamic consistency. Moreover, the allocation with commitment coincides with that of the ex ante equilibrium defined in Hanany, Klibanoff, and Mukerji (2020), which is sequentially optimal when conditional preferences are updated using the smooth rule.¹⁴

Finally, we make the following assumption to ensure that the problem is well defined.¹⁵

ASSUMPTION 2: $\gamma \geq 0$ and $\lambda \gamma \frac{\sigma_\mu^2}{\sigma_\eta^2} V(\xi_t) < 1$.

B. Subjective Beliefs and Equilibrium

We begin by defining a Nash equilibrium in our environment.

¹³We restrict attention to the set of preferences that preserve closure; namely, each member of that set remains in the set after updating. This property holds naturally for expected utility preferences with Bayesian updating.

¹⁴The conditional preference under the smooth rule is made explicit in equation (A.1) in online Appendix A.

¹⁵The first restriction, $\gamma \geq 0$, ensures that in the nonstochastic steady state under complete information, the utility function is concave in the fundamental. The second assumption stipulates that the level of ambiguity or the degree of ambiguity aversion should not exceed a certain threshold. It ensures that the ex ante objective (12) is finite for at least one strategy, so that the agent's choice set is nonempty (see the Proof of Lemma A.5 in online Appendix A for a more detailed discussion).

DEFINITION 1: A Nash equilibrium is a strategy $g(x_i^t)$ such that $k_{it} = g(x_i^t)$ maximizes the objective (12), and the aggregate outcome is consistent with individual actions, $K_t = \int g(x_i^t) di$.

We focus on linear strategies, with $g(x_i^t)$ being linear functions of the history of signals. The next proposition establishes that a linear Nash equilibrium always exists, so this focus should be seen as an equilibrium refinement rather than a restriction.¹⁶

PROPOSITION 2: A Nash equilibrium with linear strategies exists.

Without ambiguity aversion, the problem reduces to a standard beauty contest, and the optimal strategy can be written as an average of the expected fundamental and the expected aggregate outcome, as in Morris and Shin (2002). With ambiguity aversion, a similar result holds, albeit the expectations need to be based on the endogenous subjective beliefs. Consider the first-order condition for maximizing (12) with respect to the individual action k_{it} :

$$(13) \quad \int_{\mu^t} E^{\mu^t} \left[\frac{\partial u(k_{it}, K_t, \xi_t)}{\partial k_{it}} \middle| x_i^t \right] \phi'(E^{\mu^t}[u(k_{it}, K_t, \xi_t)]) p(\mu^t | x_i^t) d\mu^t = 0.$$

Notice that when evaluating the payoff implications of an action, relative to the Bayesian kernel $p(\mu^t | x_i^t)$, agents behave as if they were using an expectation kernel distorted by $\phi'(E^{\mu^t}[u(k_{it}, K_t, \xi_t)])$, which we refer to as the agents' subjective beliefs. This distortion reflects the fact that, whenever a model generates lower ex ante expected utility in equilibrium, agents would regard it as the more likely model in their posterior belief relative to the Bayesian posterior. Recall that we have already employed this line of argument when explaining the bias term in the illustrating example. In the special case in which agents are ambiguity neutral, $\phi'(\cdot) = 1$, and the subjective kernel coincides with the Bayesian one. The following proposition summarizes this discussion.

PROPOSITION 3: Taking the law of motion of K_t as given, agent i 's best response satisfies

$$(14) \quad k_{it} = (1 - \alpha) \mathcal{F}_{it}[\xi_t] + \alpha \mathcal{F}_{it}[K_t],$$

where $\mathcal{F}_{it}[\cdot]$ denotes agent i 's subjective expectation operator, that is,

$$\mathcal{F}_{it}[\cdot] \equiv \int_{\mu^t} E^{\mu^t}[\cdot | x_i^t] \hat{p}(\mu^t | x_i^t) d\mu^t,$$

and $\hat{p}(\mu^t | x_i^t) \propto \phi'(E^{\mu^t}[u(k_{it}, K_t, \xi_t)]) p(\mu^t | x_i^t)$.

¹⁶With strategic interactions, we cannot rule out the possibility that all agents collectively choose a nonlinear strategy. That being said, as shown in online Appendix D, when the coordination motive vanishes (as in Section I), there exists a unique solution to the individual's forecasting problem, which is linear in signals.

Importantly, since agents' payoffs depend on the aggregate outcome K_t , both the magnitude and dynamics of belief distortions hinge on the equilibrium coordination motive. Consequently, besides needing to anticipate the actions and beliefs of others, the way that agents form their beliefs is also affected by the aggregate outcome. The fact that the equilibrium outcome and the distorted subjective beliefs have to be jointly determined makes solving the equilibrium significantly more involved.

The equilibrium outcome can also be expressed as a weighted sum of higher-order subjective expectations.

COROLLARY 1: *In equilibrium, the aggregate outcome is a function of a weighted sum of infinite subjective higher-order expectations of ξ_t ,*

$$(15) \quad K_t = (1 - \alpha) \sum_{j=0}^{\infty} \alpha^j \bar{\mathcal{F}}_t^{j+1}[\xi_t],$$

where $\bar{\mathcal{F}}_t^1[\cdot] \equiv \int \mathcal{F}_{it}[\cdot] di$, and $\bar{\mathcal{F}}_t^{j+1}[\cdot] \equiv \int \mathcal{F}_{it}[\bar{\mathcal{F}}_t^j[\cdot]] di$.

This result indicates that when forming their own beliefs, agents must take into account the possible bias and altered sensitivity in others' forecasts, and this incentive is regulated by the degree of strategic complementarity. In Section IID, we show that this representation helps uncover the interaction between coordination and ambiguity aversion.

C. Equilibrium Characterization

In this subsection, we provide an equivalence result that circumvents the determination of higher-order subjective beliefs. We show that solving for the equilibrium strategy described above can be reduced to solving a single-agent Bayesian forecasting problem with a modified information structure. This forecasting problem can then be tackled using the Kalman filter, which facilitates the development of a convenient toolbox for solving models featuring ambiguity and persistent learning. The straightforward characterization it provides also allows us to derive valuable comparative statics results under general conditions.

Auxiliary Forecasting Problem.—First consider the following auxiliary inference problem about the fundamental with Bayesian forecasters, which we later link back to the economy with ambiguity.

DEFINITION 2: *The (w, α) -modified signal process is given by*

$$(16) \quad \tilde{\xi}_t = a(L) \tilde{\eta}_t, \text{ with } \tilde{\eta}_t \sim \mathcal{N}(0, (1 + w) \sigma_\eta^2),$$

$$(17) \quad \tilde{x}_{it} = m(L) \tilde{\eta}_t + n(L) \tilde{\epsilon}_{it}, \text{ with } \tilde{\epsilon}_{it} \sim \mathcal{N}(0, (1 - \alpha)^{-1} \Sigma),$$

where w is a nonnegative scalar and α is the degree of complementarity. Let the optimal Bayesian forecast be given by

$$(18) \quad \tilde{E}_{it}[\tilde{\xi}_t] = p(L; w, \alpha) \tilde{x}_{it}.$$

This signal process imposes two modifications relative to the original process. The volatility of the innovation to the fundamental is amplified by a factor of $(1 + w)$, and the covariance matrix of the idiosyncratic noise is amplified by $(1 - \alpha)^{-1}$. Intuitively, ambiguity aversion leads to a more diffuse prior about η_t , while the coordination motive reduces agents' incentives to rely on private information. These two considerations from the original environment are captured by the modifications to the shock processes in this auxiliary forecasting problem.

Sensitivity and Bias.—Next, we introduce a notion of aggregate sensitivity and bias in our multivariate setting, which helps characterize the equilibrium strategy.

DEFINITION 3: *Define the aggregate sensitivity to signals as*

$$(19) \quad \mathcal{S} \equiv 1 - \frac{\text{cov}[\xi_t - K_t, \xi_t]}{V(\xi_t)},$$

and the bias as

$$(20) \quad \mathcal{B} \equiv E[\xi_t] - E[K_t].$$

With multiple signals, the sensitivity to signals can no longer be measured by the loading on a particular signal. To better understand the definition in equation (19), notice that under perfect information, the aggregate outcome K_t simply mirrors the exogenous fundamental ξ_t , and $\mathcal{S} = 1$. When information is incomplete, however, agents receive noisy signals, and their aggregate response is dampened, reducing $\mathcal{S}$. Moving to the definition of bias in equation (20), notice that without ambiguity, the unconditional mean of the aggregate outcome coincides with that of the fundamental, resulting in $\mathcal{B} = 0$, even if information is incomplete. Ambiguity aversion, however, can lead to a permanent bias in agents' actions.

These definitions for sensitivity and bias can also be viewed as the counterparts of the regression coefficients in the following equation:

$$(21) \quad \xi_t - K_t = \beta_0 + \beta_1 \xi_t + \text{residuals},$$

where β_0 corresponds to the bias, $\mathcal{B}$, and β_1 corresponds to the reverse of the sensitivity, $1 - \mathcal{S}$.¹⁷ These two moments contain rich information about agents' subjective beliefs and about the expectation formation process. As in Section I, the levels of bias and sensitivity are jointly determined in equilibrium.

Equilibrium Strategy.—Despite the complex interactions between persistent information, coordination motives, and ambiguity aversion, the equilibrium strategy ultimately takes a relatively simple form. This form can be connected to the notion of aggregate sensitivity and bias. Let $\tau_\mu \equiv \sigma_\mu^2 / \sigma_\eta^2$ be a normalized measure of the amount of ambiguity.

¹⁷ When K_t stands for the consensus forecasts about ξ_t , regression (21) resembles the ones explored in Kohlhas and Walther (2021).

PROPOSITION 4: *The linear strategy in equilibrium takes the following form:*

$$(22) \quad g(x_t^i) = (1 + r)p(L; w, \alpha)x_{it} + \mathcal{B}.$$

(i) *The polynomial matrix $p(L; w, \alpha)$ is the Bayesian forecasting rule in (18) with the (w, α) -modified signal process, and w satisfies*

$$(23) \quad w = \left[\frac{1}{\tau_\mu} - \lambda(1 - \alpha) \left(V(\xi_t - K_t) + \frac{\lambda\gamma\tau_\mu V(\xi_t)^2(1 - \mathcal{S})^2}{1 - \lambda\gamma\tau_\mu V(\xi_t)} \right) \right]^{-1} \geq \tau_\mu.$$

(ii) *The additional amplification, r , satisfies*

$$(24) \quad r = \gamma \frac{\lambda\tau_\mu V(\xi_t)}{1 - \lambda\gamma\tau_\mu V(\xi_t)} \frac{w}{1 + w} (1 - \mathcal{S}) \geq 0.$$

(iii) *The level of bias, $\mathcal{B}$, satisfies*

$$(25) \quad \mathcal{B} = \chi \frac{\lambda\tau_\mu V(\xi_t)}{1 - \lambda\gamma\tau_\mu V(\xi_t)} (1 - \mathcal{S}).$$

To understand this result, let us first ignore ambiguity ($\tau_\mu = 0$). In this case, the equilibrium strategy reduces to $p(L; 0, \alpha)$. This means agents' actions are equivalent to forming Bayesian expectations about ξ_t , with the volatility of idiosyncratic noise adjusted to accommodate coordination considerations. This equilibrium characterization bypasses the dependence on higher-order beliefs and includes the single-agent results from Huo and Pedroni (2020).

When ambiguity is present ($\tau_\mu > 0$), prior uncertainty looms larger. This effect is captured by the amplified variance of the fundamental, by a factor of $(1 + w)$, in equation (16). Following the new forecasting rule $p(L; w, \alpha)$, agents' reliance on their signals increases when making forecasts. This channel makes it appear as if agents are overconfident in their signals, as in Broer and Kohlhas (2024). Moreover, remember that the term $-\chi\xi_t - \frac{1}{2}\gamma\xi_t^2$ in the utility function captures the nonstrategic impact of the fundamental on the agent's payoff. A positive γ introduces an additional reason for agents to react to signals: the dependence on ξ_t^2 intensifies the impact of extreme realizations of the prior mean μ_t on welfare. This pushes agents' actions further away from the Bayesian benchmark through its effect on r . This additional overreaction is similar to the diagnostic expectations explored in Bordalo et al. (2020), but in our environment the magnitude of this overreaction is endogenously determined in equilibrium.¹⁸

Additionally, if $\chi \neq 0$, the equilibrium strategy exhibits a permanent bias. Keep in mind that agents' subjective beliefs, $\mathcal{F}_{it}[\cdot]$, assign more weight to models that generate lower ex ante expected utility, leading to a pessimistic bias.¹⁹ Condition

¹⁸It should be noted that the implied overreaction in our equilibrium is built on a different microfoundation than those in Broer and Kohlhas (2024) and Bordalo et al. (2020). These approaches are suitable for different applied settings and are not necessarily substitutes.

¹⁹Notice that the sign of the bias depends on the sign of χ .

(25) indicates that the magnitude of the bias $\mathcal{B}$ and the level of sensitivity $\mathcal{S}$ are endogenously connected and constrained by equilibrium conditions. It is worth noting that condition (25) does not imply a direct inverse relationship between sensitivity and bias, as both are ultimately functions of deep parameters in the model. For instance, in Section I, we observe that when the degree of ambiguity aversion varies, both bias and sensitivity move in the same direction. The joint pattern of bias and sensitivity yields testable implications, which we explore in Section III.

Computation.—Even without ambiguity, solving for the equilibrium presents a significant challenge due to the combination of persistent information and coordination motives. Agents' strategies are infinite-dimensional objects with the entire history of signals as state variables. Although a finite-state representation is possible when signals follow finite ARMA (p, q) processes (Woodford 2003; Angeletos and La'O 2010; Huo and Pedroni 2020), it remains unclear whether these results could be extended to models with ambiguity aversion. The following corollary confirms they can.

COROLLARY 2: *When the fundamental and signals follow finite ARMA processes, the equilibrium strategy admits a finite-state representation.*

Proposition 4 shows that the finite-state representation of the equilibrium can be achieved with ambiguity-averse agents, persistent information, and general equilibrium considerations. It also outlines a clear computation method:²⁰

- For a particular pair of (w, α) , the Bayesian forecasting rule $p(L; w, \alpha)$ can be obtained using standard algorithms such as the Kalman filter.
- The value of r can be determined using condition (24). Together with the equilibrium strategy (22), this leads to the outcome K_t and the sensitivity $\mathcal{S}$. At this stage, the bias can be ignored, as it does not affect any of the terms in the formulas.
- Condition (23) can then be used to iterate on the value of w until convergence.
- Finally, the bias can be obtained from equation (25).

In summary, this complex, seemingly infinite-dimensional problem effectively reduces to a one-dimensional fixed-point problem about w . In Section III, we leverage this result to characterize the dynamics of inflation forecasts under ambiguity, which boils down to solving a single cubic equation.

²⁰We could also accommodate the case where the signals contain endogenous aggregate variables. Since each agent behaves in a competitive way, they still treat the information as exogenous even if signals are endogenous. Therefore, adding endogenous signals amounts to adding another layer of fixed point problem on top of our equivalence result.

D. Role of General Equilibrium Considerations

How do GE considerations interact with ambiguity aversion? To answer this question, it is useful to revisit the higher-order expectation representation in equation (15). A change in α not only affects the responses of expectations at each order but also shifts the relative weight assigned to each expectation. However, these intensive and extensive margins do not necessarily move in the same direction in shaping equilibrium outcomes.

To illustrate the intuition, we start with a simple example with static information and then extend the findings to general information processes. Suppose that the utility function is given by

$$u(k_i, K, \xi) = -\frac{1}{2}[(1 - \alpha)(k_i - \xi)^2 + \alpha(k_i - K)^2] - \chi\xi$$

and that agents perceive ambiguity about the fundamental according to

$$\xi \sim \mathcal{N}(\mu, \sigma_\xi^2), \text{ and } \mu \sim \mathcal{N}(0, \sigma_\mu^2),$$

where μ is objectively zero. Agents also observe a private signal $x_i \sim \mathcal{N}(\xi, \sigma_\epsilon^2)$. In this simple setup, we can further evaluate the effects of GE considerations on sensitivity and bias, by examining the properties of subjective higher-order expectations.

PROPOSITION 5: *In the example economy, subjective higher-order expectations obey the following structure.*

(i) *The m -th order subjective expectation defined in condition (15) is given by*

$$\bar{\mathcal{F}}^m[\xi] = \kappa_m \xi + \beta_m,$$

$$\text{where } \kappa_m = \left(\frac{(1+w)\sigma_\xi^2}{(1+w)\sigma_\xi^2 + \sigma_\epsilon^2} \right)^m \text{ and } \beta_m = \beta_{m-1} + (\kappa_m - \kappa_{m-1})\lambda\chi\sigma_\mu^2.$$

(ii) *The endogenous multiplier w is not monotonic in α .*

(iii) *The aggregate outcome is given by*

$$K = S\xi + B = (1 - \alpha) \sum_{m=0}^{\infty} \alpha^m \kappa_m \xi + (1 - \alpha) \sum_{m=0}^{\infty} \alpha^m \beta_m.$$

Part (i) implies that for a fixed α , as the order increases, the sensitivity of the subjective higher-order expectations, κ_m , decreases with the order. This is displayed in panel A of Figure 1 and resembles the rational expectations result without ambiguity (Morris and Shin 2002). Interestingly, panel B of Figure 1 shows that the bias, β_m , is increasing in the order of expectation. This is a result of the accumulation of pessimism in ascending the hierarchy of beliefs, a footprint of ambiguity aversion.

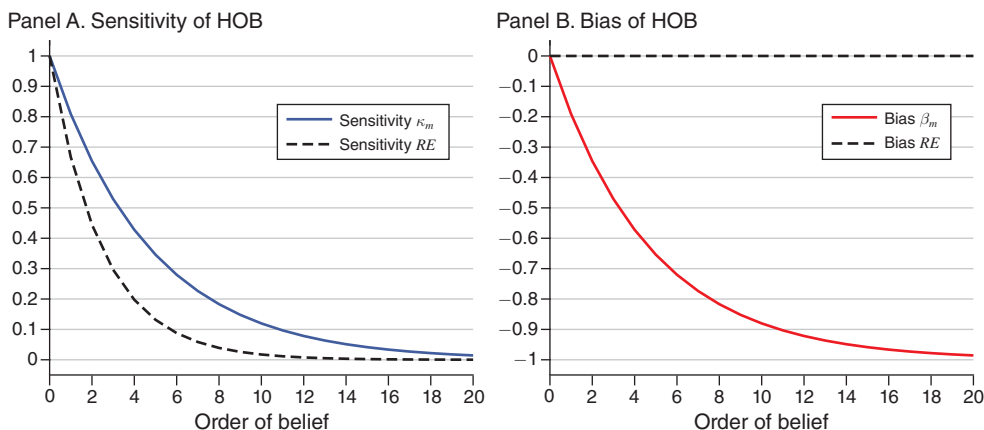


FIGURE 1. SUBJECTIVE HIGHER-ORDER BELIEFS

Note: This figure reports sensitivity (panel A) and bias (panel B) associated with the entire belief hierarchy as a function of the order of beliefs.

When forecasting the beliefs of others, agents internalize the bias of others in addition to their own.

Part (ii) suggests that there are competing forces shaping the effect of a change in α on w . *Ceteris paribus*, a higher α shifts agents' attention from forecasting the fundamental to forecasting others' actions. As a result, private information about the fundamental becomes less relevant, and agents reduce how much they respond to signals. As can be seen in condition (23), the endogenous amplification force captured by w vanishes when α approaches 1.

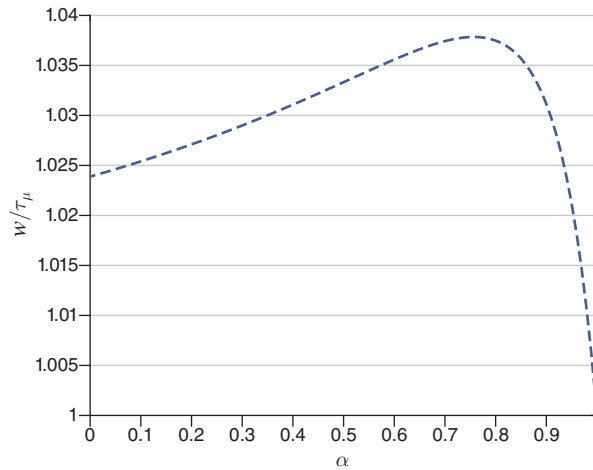
However, this is not the only force at play. Condition (23) also reveals that a reduction in the sensitivity S could contribute to a higher w . This is due to the fact that, when agents reduce the responsiveness to signals, they also increase their reliance on their ambiguous prior, and this is costly since agents are ambiguity averse. Endogenously, this puts upward pressure on w to undo this effect. Overall, these two forces leave the comparative statics of w with respect to α ambiguous. As shown in Figure 2, w exhibits a humped shape with respect to the strength of GE considerations.

This nonmonotonicity is inherited by both κ_m and β_m . Together with Part (i) of Proposition 5, this implies that sensitivities and biases at every order mirror the pattern of w and are not monotonic with respect to the GE consideration α .

Part (iii) states that the observed sensitivity and bias of the aggregate outcome is a weighted average of κ_m and β_m . Mechanically, as α increases, a larger weight is assigned to higher-order expectations relative to first-order expectations.

Ultimately, when α changes, the reweighting channel dominates the ambiguous effects on w , leading to overall lower sensitivity and increased bias. These observations hold beyond this simple example. Taking advantage of Proposition 4, we obtain the following result for a general information structure.

PROPOSITION 6: *When $\gamma = 0$, the sensitivity S is decreasing in α , and the magnitude of bias $|\mathcal{B}|$ is increasing in α .*

FIGURE 2. COMPARATIVE STATICS OF w OVER α

E. Extensions

The results discussed so far depend on assumptions about the payoff function and the information structure. However, the observational equivalence result and the insights regarding bias and sensitivity are more general.

Multiple Actions.—In the baseline specification, each agent chooses only one action, while firms and households often need to simultaneously consider multiple choices. For example, a manager must jointly plan hiring and advertisement expenditures; a consumer must decide on the allocation of time for work-related activities, educational activities, household work, and so on. We now extend our baseline specification to accommodate this possibility.

We maintain the specification for the shocks and ambiguity but modify the preference toward risk as follows:

$$(26) \quad u(k_{it}, K_t, \xi_t) = \frac{1}{2}(k_{it} - \kappa \xi_t)' \Psi_k (k_{it} - \kappa \xi_t) + \frac{1}{2}(k_{it} - K_t)' \Psi_K (k_{it} - K_t) + \chi \xi_t - \frac{1}{2} \gamma \xi_t^2,$$

which is a generalization of the univariate case (11) to allow for multiple actions.²¹ Here, the individual actions, k_{it} , and the aggregate actions, K_t , are both $J \times 1$ random vectors. The constant $J \times J$ matrices Ψ_k and Ψ_K contain information on how individual and aggregate actions affect welfare in the steady state. Finally, κ is a $J \times 1$ vector of constants that determines how agents would respond to the fundamental in a full information economy.

Unlike the baseline environment, an individual agent's best response leads to a multivariate beauty contest game.

²¹This specification is derived from a quadratic approximation of generic preferences, which is efficient under incomplete information. See online Appendix B.1 for details.

LEMMA 1: Denote subjective expectations by $\mathcal{F}_{it}[\cdot]$ as in Proposition 3. The best response satisfies

$$k_{it} = (I - \Theta) \mathcal{F}_{it}[\kappa \xi_t] + \Theta \mathcal{F}_{it}[K_t], \text{ with } \Theta \equiv (\Psi_k + \Psi_K)^{-1} \Psi_K.$$

In addition, denote the eigenvalue decomposition of Θ as

$$\Theta = Q^{-1} \text{diag}(\alpha_1, \dots, \alpha_J) Q.$$

The best-response system can be transformed into

$$Qk_{it} = [I - \text{diag}(\alpha_1, \dots, \alpha_J)] \mathcal{F}_{it}[Q\kappa \xi_t] + \text{diag}(\alpha_1, \dots, \alpha_J) \mathcal{F}_{it}[QK_t].$$

With multiple actions, the interdependence between actions makes strategic complementarities more involved, summarized now by the matrix Θ . When information is perfect and there is no ambiguity about the fundamental, $k_{it} = K_t = \kappa \xi_t$, and the matrix Θ is irrelevant. When information is incomplete, Θ starts to play a role.

The second part of Lemma 1 shows that the multivariate system can be transformed into a seemingly decoupled system: the vector of transformed actions, Qk_{it} , is orthogonal in the sense that the corresponding complementarities are now represented by a diagonal matrix. This is indeed the case when the expectation operator is independent of agents' actions. However, when agents are ambiguity averse, subjective beliefs depend on expected utility. As a result, the actions remain interconnected through the endogeneity of the beliefs.

To proceed, we define bias and sensitivity for the multiple action case, now given by $J \times 1$ vectors,

$$\mathcal{B} \equiv E[\kappa \xi_t] - E[K_t], \text{ and } \mathcal{S} \equiv \mathbf{1} - \frac{\text{cov}[\kappa \xi_t - K_t, \xi_t]}{V(\xi_t)}.$$

We also introduce the $J \times J$ matrix W that generalizes the scalar w from condition (23),

$$W \equiv \left[\tau_\mu^{-1} I + \lambda Q \left(V(\kappa \xi_t - K_t) + \frac{\lambda \gamma \tau_\mu V(\xi_t)^2}{1 - \lambda \gamma \tau_\mu V(\xi_t)} (\mathbf{1} - \mathcal{S})(\mathbf{1} - \mathcal{S})' \right) \Psi_k Q^{-1} \right]^{-1}.$$

The following proposition circumvents the aforementioned difficulties and establishes an equivalence result that resembles the univariate one.

PROPOSITION 7: The linear strategy in equilibrium takes the following form:²²

$$k_{it} = g(x_i^t) = Q^{-1} P^{-1} \begin{bmatrix} \left(\sum_{j=1}^J P_{1j} \left[1 + \frac{(1-\alpha_j)}{(1-\alpha_1)} r_j \right] e_j' Q \kappa \right) p(L; w_1, \alpha_1) \\ \vdots \\ \left(\sum_{j=1}^J P_{Jj} \left[1 + \frac{(1-\alpha_j)}{(1-\alpha_J)} r_j \right] e_j' Q \kappa \right) p(L; w_J, \alpha_J) \end{bmatrix} x_{it} + \mathcal{B}.$$

²²The vector e_j denotes the j -th column of the $J \times J$ identity matrix.

- (i) The polynomial matrix $p(L; w_j, \alpha_j)$ is the Bayesian forecasting rule in (18). The matrix P and the scalars w_j derive from the following eigenvalue decomposition:

$$[I - \text{diag}(\alpha_1, \dots, \alpha_J)](\tau_\mu^{-1}I + W) = P^{-1} \text{diag}(\omega_1, \dots, \omega_J)P,$$

$$\text{and } w_j \equiv \frac{\tau_\mu \omega_j}{1 - \alpha_j} - 1.$$

- (ii) The additional amplification, r_j , satisfies

$$r_j = \gamma \frac{\lambda \tau_\mu V(\xi_t)}{1 - \lambda \gamma \tau_\mu V(\xi_t)} \frac{e_j' W Q (\mathbf{1} - \mathcal{S})}{e_j' Q \kappa_1 (1 + w_j)}.$$

- (iii) The bias vector, $\mathcal{B}$, satisfies

$$\mathcal{B} = \chi \frac{\lambda \tau_\mu V(\xi_t)}{1 - \lambda \gamma \tau_\mu V(\xi_t)} (\mathbf{1} - \mathcal{S}).$$

The equilibrium strategy is effectively a linear combination of J different Bayesian forecasts of the fundamental. In these modified forecasts, the idiosyncratic noises are adjusted according to the eigenvalues of the matrix of complementarities, Θ , which is exogenous. On the other hand, the shocks to fundamentals are adjusted according to the eigenvalue of an endogenous matrix involving W . The matrix P provides the required rotation, on top of the matrix Q , so that the vector of actions can be analyzed in an orthogonal way, even when ambiguity aversion is present. Notably, the biases for each action share a common component but also depend on their specific correlation with the fundamental, captured by the vector κ . This allows the bias to vary in direction for different actions.

Other Extensions.—In online Appendix B.2, we allow the economy to be inefficient under both complete and incomplete information. Essentially, our equivalence result applies to general quadratic payoff structures. Our optimal-policy application in Section IV utilizes this set of results.

In another extension, online Appendix B.3 shows how to generalize these insights to settings with multiple aggregate shocks, including common noises. This extension can be used to study the interaction between ambiguity and the effects of nonfundamental driven fluctuations, for example.

Finally, online Appendix F explores the effects of ambiguity on the value of increasing signal precision. The answer to this question is particularly relevant for models of rational inattention, where signal precision is chosen endogenously, balancing the cost and benefit of acquiring information. We show that the value of higher precision in signals is increasing in the amount of ambiguity.

III. Application: Inflation Forecasts

Under full information rational expectations (FIRE), optimal forecasts are unbiased, and forecast errors are serially uncorrelated. In contrast, using survey data on

expectations, the literature has demonstrated that the average forecasts of agents are biased and that forecast errors are serially correlated.²³ In this section, we document additional survey evidence on the joint behavior of bias and persistence across the income distribution, which cannot be easily rationalized by existing theories of expectation formation. We then show that these facts emerge naturally from a microfounded model with ambiguity-averse consumers.

A. Data and Facts

The Michigan Survey of Consumers collects data on household inflation expectations, asking what is their “price expectations for the next 12 months.” It also provides information about household income, which allows us to allocate surveyed households in each quarter into groups based on their income percentiles. For each income group g , the average inflation forecast error in quarter t is calculated as the average inflation forecast error across every household $i \in \mathbb{I}_g$:

$$\overline{FE}_{g,t} \equiv \int_{\mathbb{I}_g} (\pi_t - \mathcal{F}_{i,g,t}[\pi_t]) di,$$

where we continue to use $\mathcal{F}_{i,g,t}$ to denote households’ subjective expectations. The bias and persistence of forecast errors for each income group are given by their across-time average and autocorrelation:

$$(27) \quad Bias_g \equiv \frac{1}{T} \sum_{t=1}^T \overline{FE}_{g,t}, \text{ and } Persistence_g \equiv \text{corr}(\overline{FE}_{g,t}, \overline{FE}_{g,t-1}).$$

Figure 3 presents the bias (panel A) and persistence (panel B) of households’ one-year-ahead inflation forecasts. In line with the existing literature, households’ inflation forecasts are biased upward, leading to negative average forecast errors. What is interesting is the joint behavior of bias and persistence in the cross section of household incomes. As households move up the income ladder, the bias in their inflation expectations decreases, while the persistence of their inflation forecast errors increases.

In online Appendix H.1, we further document several additional results. (i) In the Survey of Consumer Expectations, conducted by the Federal Reserve of New York, the joint pattern of bias and persistence of forecast errors is similar to the one observed in the MSC, as shown in online Appendix Table H.1. (ii) At the individual level, after controlling for other observed characteristics such as age and resident state, the magnitude of the bias continues to decrease with income, as shown in online Appendix Table H.2.

The documented patterns of bias and persistence of forecast errors present challenges to the assumption of rational expectations. Within the rational expectations paradigm, whether under full or noisy information, forecast errors should be zero on average. Noisy rational expectations can generate persistent forecast errors, but

²³ See, for example, Farmer, Nakamura, and Steinsson (2024) for evidence of bias and autocorrelated forecast errors in the survey of professional forecasters; Kohlhas and Robertson (2024) for evidence of bias in professional forecasters’ forecasts; and Bhandari, Borovička, and Ho (2024) for evidence of bias in household forecasts.

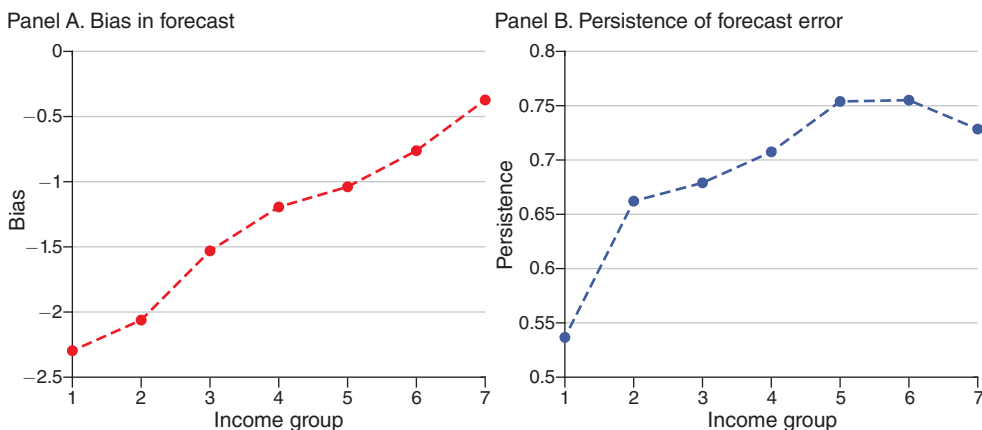


FIGURE 3. BIAS AND PERSISTENCE OF FORECAST ERROR IN THE SURVEY DATA

Notes: This figure reports bias (panel A) and persistence (panel B) of households' inflation forecasts in the cross section of the income distribution. Bias and persistence of each income percentile are calculated by the mean and serial correlation of forecast errors of households' inflation expectations for the next 12 months. Data are obtained from the University of Michigan, Survey Research Center, Surveys of Consumers (1987:I–2020:IV) and US CPI Inflation (1987:IV–2021:III).

matching the increasing pattern of persistence across household income would require having richer households being less informed about the economy. A notable exception is Farmer, Nakamura, and Steinsson (2024). They demonstrate that slow learning over the long-run trend with a unit root can create bias and persistence in forecast errors. However, the question of why bias and persistence move in opposite directions as household income changes remains unaddressed.

In what follows, we set up a two-period consumption model with sticky nominal income and ambiguity about the inflation process. Applying the results developed in Section II, we show that the inflation forecasts that arise from this model are consistent with the type of cross-sectional distribution of bias and persistence documented above.

B. Model

Household Problem.—There is a finite number of household groups indexed by $g \in 1, \dots, N$, each differing in their nominal income level, denoted by Y_g . Within each income group, there is a continuum of households indexed by i . We consider a simple, stylized consumption-saving problem where households plan their consumption only for periods t and $t + 1$. The utility function of the household is given by

$$U(C_{i,g,t}, C_{i,g,t+1}) = \frac{C_{i,g,t}^{1-\nu} - 1}{1-\nu} + \beta \frac{C_{i,g,t+1}^{1-\nu} - 1}{1-\nu},$$

where ν controls the degree of risk aversion of households.

Nominal income is rigid between t and $t + 1$. A household in group g receives nominal income $P_t Y_g / 2$ in each period. Therefore, their budget constraint is given by

$$P_t C_{i,g,t} + P_{t+1} C_{i,g,t+1} = P_t Y_g.$$

Let $\pi_{t+1} \equiv (P_{t+1} - P_t)/P_t$ denote the inflation rate. The budget constraint for household i can then be rewritten as

$$C_{i,g,t} + (1 + \pi_{t+1})C_{i,g,t+1} = Y_g.$$

That is, a higher inflation rate makes consumption tomorrow more expensive relative to consumption today, adversely affecting households due to the nominal rigidity in their income.²⁴ In this problem, households only face uncertainty about the inflation rate. Once the belief about future inflation is determined, the optimal consumption plan follows. The following lemma directs us to concentrate on inflation expectations.

LEMMA 2: *Around the zero-inflation steady state:*

- (i) *The optimal consumption change, $c_{i,g,t}$, is proportional to the household's subjective expectation about inflation*

$$(28) \quad c_{i,g,t} = \frac{\beta^{1/\nu}}{1 + \beta^{1/\nu}} \mathcal{F}_{i,g,t}[\pi_{t+1}].$$

- (ii) *The quadratic approximation of the utility function $U(C_{i,g,t}, C_{i,g,t+1}) \approx Q(\mathcal{F}_{i,g,t}[\pi_{t+1}], \pi_{t+1})$ is given by*

$$(29) \quad Q(\mathcal{F}_{i,g,t}[\pi_{t+1}], \pi_{t+1}) = -\delta_g (\mathcal{F}_{i,g,t}[\pi_{t+1}] - \pi_{t+1})^2 - \chi_g \pi_{t+1} - \gamma_g \pi_{t+1}^2,$$

where δ_g , χ_g , and γ_g are positive, when $\nu > 1$, and satisfy

$$\delta_g, \chi_g, \gamma_g \propto Y_g^{1-\nu}.$$

The first part of Lemma 2 states that there is a one-to-one mapping between the optimal choice of consumption and the household's subjective expectation about the inflation rate. A higher inflation expectation implies that future goods become more expensive relative to current goods, and households have an incentive to increase their current consumption.

Condition (28) allows us to express the utility in terms of expected inflation and actual inflation since future consumption depends on the realized inflation. This quadratic approximation of the utility function is nested within the general specification (11) in Section IIA. The utility function (29) reveals two important properties that are crucial for matching the data later on: (i) *ceteris paribus*, higher inflation lowers household welfare since both χ_g and γ_g are positive; (ii) a higher level of income reduces households' exposure to variations in inflation since $(\delta_g, \chi_g, \gamma_g)$ are decreasing in Y_g .

²⁴The assumption of complete nominal income rigidity is made only for simplicity. What is important is that, relative to goods prices, nominal income changes at a slower rate. Implicitly, this model setup assumes that the central bank targets nominal income, the income shares in different groups are invariant to business cycles, and business cycles are mostly driven by supply-side shocks that directly affect the inflation rate.

Subjective Expectations.—The inflation rate, π_t , follows an exogenous AR(1) process,

$$\pi_t = \rho\pi_{t-1} + \eta_t, \text{ with } \eta_t \sim \mathcal{N}(0, \sigma_\eta^2),$$

and at each period t , household i receives a noisy private signal,

$$x_{i,g,t} = \pi_t + \epsilon_{i,g,t}, \text{ with } \epsilon_{i,g,t} \sim \mathcal{N}(0, \sigma_\epsilon^2).$$

The information set of household i at time t is $\mathcal{I}_{i,g,t} = \{x_{i,g,t}, x_{i,g,t-1}, \dots\}$. So far, this specification is similar to the structure used in the literature that studies survey evidence on inflation forecasts (Coibion and Gorodnichenko 2015; Bordalo et al. 2020). It is also worth noting that the received information is independent of the household's income level.

We depart from rational expectations by allowing agents to perceive ambiguity about the mean of the innovation to inflation,

$$\eta_t \sim \mathcal{N}(\mu_t, \sigma_\eta^2), \text{ and } \mu_t \sim \mathcal{N}(0, \sigma_\mu^2),$$

where, again, σ_μ^2 corresponds to the amount of ambiguity. Let $h(x_{i,g}^t)$ denote the households' strategy in forming expectations, that is, $\mathcal{F}_{i,g,t}[\pi_{t+1}] = h(x_{i,g}^t)$. The mapping from subjective inflation expectations to optimal consumption in period t allows us to transform the original optimal consumption problem into an optimal forecasting problem that can be embedded into our general theoretical framework. Following the specification from Section II, households maximize the following objective function:

$$(30) \quad \phi^{-1} \left(\int_{\mu^t} \phi \left(E_{i,g,t}^{\mu^t} [\mathcal{Q}(h(x^t), \pi_{t+1})] \right) p(\mu^t) d\mu^t \right).$$

By establishing the equivalence between the optimal forecasting problem (30) and the original optimal consumption problem, we are implicitly assuming that households respond to the survey questions about inflation by reporting beliefs that are consistent with their consumption decisions. This assumption is common in studies that connect survey data on expectations to structural models with ambiguity-averse agents, such as Bhandari, Borovička, and Ho (2024) and Pei (2024).

C. Inflation Forecasts under Ambiguity

Bayesian Benchmark.—We begin by muting ambiguity ($\sigma_\mu^2 = 0$) and considering the predictions of this model under standard Bayesian expectations, characterized in the following proposition.

PROPOSITION 8: *The Bayesian forecast is a weighted sum of the prior and the new signal,*

$$(31) \quad E_{i,g,t}[\pi_{t+1}] = \omega E_{i,g,t-1}[\pi_t] + (\rho - \omega)x_{i,g,t},$$

and the average forecast error satisfies

$$\pi_{t+1} - \bar{E}_{g,t}[\pi_{t+1}] = \frac{1}{1 - \omega_L} \eta_{t+1},$$

where the persistence ω is given by

$$\omega = \frac{1}{2} \left[\rho + \frac{\sigma_\epsilon^2 + \sigma_\eta^2}{\rho \sigma_\epsilon^2} - \sqrt{\left(\rho + \frac{\sigma_\epsilon^2 + \sigma_\eta^2}{\rho \sigma_\epsilon^2} \right)^2 - 4} \right].$$

Without ambiguity, households underreact to their signals due solely to the noise in their observations. As a result, the aggregate forecast errors are persistent over time, as discussed in Coibion and Gorodnichenko (2015). However, the persistence ω does not correlate with household income levels. Moreover, the unconditional mean of household forecasts coincides with that of actual inflation, leaving no room for permanent bias. These properties are inconsistent with the empirical facts documented in Section IIIA.

Ambiguity-Averse Households.—When households are ambiguity averse, their income level matters for their subjective beliefs. The following characterization directly applies the results developed in Section II.

PROPOSITION 9: *With ambiguity aversion, the individual subjective forecast satisfies*

$$(32) \quad \mathcal{F}_{i,g,t}[\pi_{t+1}] = \vartheta_g \mathcal{F}_{i,g,t-1}[\pi_t] + (1 + r_g)(\rho - \vartheta_g)x_{i,g,t} + (1 - \vartheta_g)\mathcal{B}_g,$$

and the average forecast error obeys

$$\pi_{t+1} - \bar{\mathcal{F}}_{g,t}[\pi_{t+1}] = \frac{1 + r_g}{1 - \vartheta_g L} \eta_{t+1} - \frac{r_g}{1 - \rho L} \eta_{t+1} - \mathcal{B}_g,$$

where the persistence ϑ_g is given by

$$(33) \quad \vartheta_g = \frac{1}{2} \left[\rho + \frac{(1 + w_g)\sigma_\eta^2 + \sigma_\epsilon^2}{\rho \sigma_\epsilon^2} - \sqrt{\left(\rho + \frac{(1 + w_g)\sigma_\eta^2 + \sigma_\epsilon^2}{\rho \sigma_\epsilon^2} \right)^2 - 4} \right] < \omega,$$

with $r_g > 0$, $w_g > 0$, and $\mathcal{B}_g > 0$.

Condition (32) shows that the subjective inflation expectation follows a law of motion similar to the Bayesian one but with several important modifications. First, households consistently overestimate the inflation rate by an amount determined by the bias term $\mathcal{B}_g$.

Second, the persistence of forecast errors is smaller than in the Bayesian case, $\vartheta_g < \omega$, for all g . Households react to their signals as if they are overconfident à la Broer and Kohlhas (2024), as the perceived signal-to-noise ratio $(1 + w_g)\sigma_\eta^2/\sigma_\epsilon^2$

is larger. Thus, households rely less on their prior means, which reduces the persistence of their forecasts.

Third, households exhibit an additional overreaction to their current signals relative to a Bayesian rule, captured by the term $1 + r_g$. The forecasting rule (32) shares similar properties to the one under diagnostic expectations (Bordalo et al. 2020) in which forecasters overweight representative states. This additional response to the signal distinguishes it from a Bayesian rule.

In the setup with ambiguity, the magnitudes of overreaction and bias endogenously depend on how inflation enters households' payoff functions. As emphasized in Lemma 2, the parameters governing a household's exposure to inflation decrease with their income level. This reduced exposure is isomorphic to a lower degree of ambiguity aversion. As a result, higher-income households are effectively less concerned about ambiguity, and their forecasting strategy is more aligned with the Bayesian benchmark. It follows that, qualitatively, richer households are less biased and their forecast errors are more persistent, which is consistent with the patterns observed in the data.

D. Results

To bring the model to the data, we set $\rho = 0.88$ and $\sigma_\eta = 0.72$ to fit the actual inflation process and fix the household's discount factor to $\beta = 0.99$.²⁵ We normalize average income to 1 and set Y_g to match the share of income of each group in the MSC. The remaining parameters are the standard deviation of private information, σ_e , the amount of ambiguity, σ_μ , the degree of ambiguity aversion, λ , and the degree of relative risk aversion, ν . We calibrate these parameters to match the persistence and bias of inflation forecast errors displayed in Figure 3. Operationally, we minimize

$$\sum_g \left(\text{Bias}_g^{\text{data}} - \text{Bias}_g^{\text{model}} \right)^2 + \left(\text{Persistence}_g^{\text{data}} - \text{Persistence}_g^{\text{model}} \right)^2.$$

The bias and persistence in the data are computed from the MSC using equation (27), and their model counterparts are the theoretical moments derived from Proposition 9.

Table 1 presents the calibrated parameters. The model fit requires both signal noise and perceived ambiguity. Figure 4 displays the goodness of fit of our calibrated model. Given its rather parsimonious structure, our model captures the cross-sectional patterns of bias and persistence of inflation forecast errors reasonably well: richer households tend to exhibit less bias in their inflation forecasts but more persistent forecast errors.

It is challenging to capture these cross-sectional patterns in the Bayesian-expectation model without ambiguity aversion, where bias is zero and persistence does not vary with income. If households perceive ambiguity ($\sigma_\mu > 0$) but their preference is ambiguity neutral ($\lambda = 0$), the persistence of forecast errors would increase with income, but bias would still not be present.

²⁵ Note that the biases of inflation forecasts in the data and in the model prediction are relative to the mean level of the actual inflation rate.

TABLE 1—CALIBRATED PARAMETERS

Param.	Value	Related to
σ_ϵ	4.11	SD of noise in private signals
σ_μ	0.82	Amount of ambiguity
λ	0.23	Degree of ambiguity aversion
ν	1.46	Risk aversion

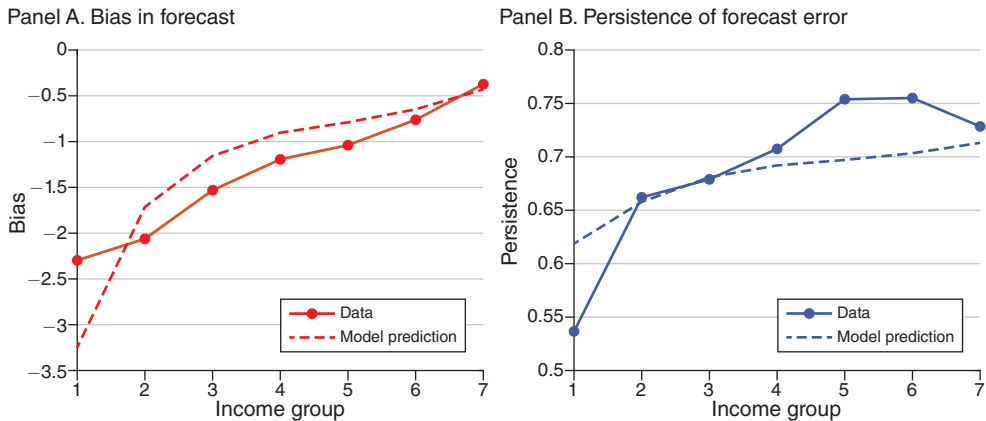


FIGURE 4. GOODNESS OF FIT

Connection with Existing Survey Evidence.—The predictions of our model are also broadly consistent with existing survey evidence on expectations. Due to dispersed noisy information, the consensus forecast underreacts to new information, implying a positive correlation between forecast errors and forecast revisions, as documented in Coibion and Gorodnichenko (2015)—henceforth, CG—using the SPF. At the individual level, this correlation would necessarily be equal to zero under rational expectations. Bordalo et al. (2020)—henceforth, BGMS—show that, in the SPF, forecasters tend to overreact to their signals, resulting in a negative correlation at the individual level.²⁶ Panel A of Figure 5 displays our model’s prediction for these two coefficients across different income groups, which are consistent with the existing evidence. Notably, the CG coefficient increases with household income, while the BGMS coefficient decreases in magnitude, which are the footprints of ambiguity aversion. In online Appendix H.2, we discuss the empirical counterparts of the CG and BGMS regression coefficients in the MSC and the SCE. The direction

²⁶To detect underreaction at the consensus level and overreaction at the individual level, we run group-specific CG and BGMS regressions of forecast errors on forecast revisions,

$$\text{CG: } \pi_{t+1} - \bar{E}_t[\pi_{t+1}] = \beta_{CG}(\bar{E}_t[\pi_{t+1}] - \bar{E}_{t-1}[\pi_{t+1}]) + \epsilon_{t+1},$$

$$\text{BGMS: } \pi_{t+1} - E_{it}[\pi_{t+1}] = \beta_{BGMS}(E_{it}[\pi_{t+1}] - E_{it-1}[\pi_{t+1}]) + \epsilon_{it+1},$$

using artificial datasets simulated by our model.

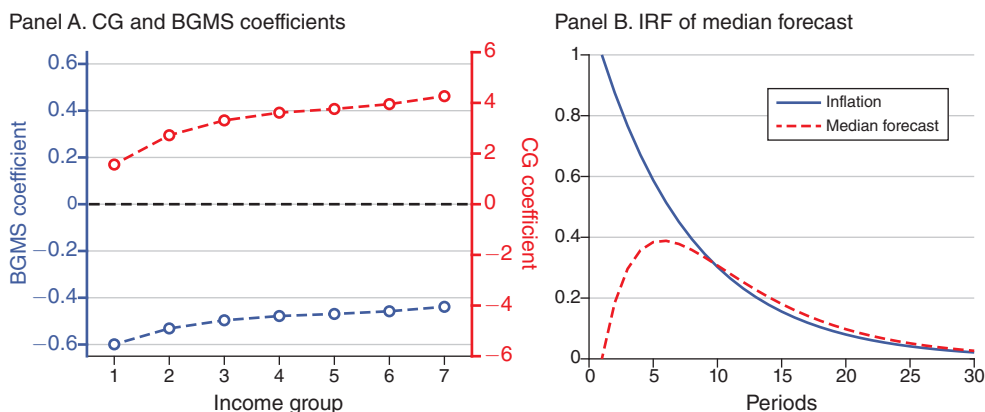


FIGURE 5. CONDITIONAL AND UNCONDITIONAL MOMENTS OF SUBJECTIVE BELIEFS

Notes: Panel A reports the theoretical CG and BGMS regression coefficients for each income group. Panel B reports the theoretical impulse response of the consensus inflation expectation for the median income group.

of change in the coefficients, as a function of income, is broadly consistent with the model predictions, though the magnitudes are not directly comparable.²⁷

Panel B shows the response of the average inflation forecast for the median income group to an inflationary shock over time. Due to the additional overreaction, associated with $1 + r_g$, the average forecast actually overshoots the true inflation rate. The implied sign-switching pattern is consistent with the empirical findings in Angeletos, Huo, and Sastry (2021).²⁸

Robustness and Alternatives.—In our model, we focus on the effect of inflation on labor income, thereby abstracting from the balance sheet effects, which Doepke and Schneider (2006) have documented could be more detrimental to richer households. In online Appendix H.3, we argue, using data from the Survey of Consumer Finances, that these effects are unlikely to dominate the labor-income effects, especially for the bottom four quintiles of the income distribution, for whom capital and business income account for less than 7 percent of their total income. The claim that poorer households are more affected by inflation is also supported by the findings in Cao et al. (2021), who compute the welfare effects of inflation across the distribution in a quantitative general equilibrium model. We also document, using the Census Bureau's House Purse Survey, that poorer households consistently report higher levels of inflation concern. This is particularly relevant to our analysis, as the mechanism hinges directly on households' subjective perceptions of inflation. These last points are also reassuring against potential general equilibrium effects overturning the direction of inflation effects.²⁹

²⁷ Due to limitations of the MSC and SCE data, we cannot exactly construct the term structure of the forecasts. In online Appendix H.2, we provide an approximate version of these regression coefficients.

²⁸ Note that in the IRF, we do not include the bias term. When conducting the projection method in Angeletos, Huo, and Sastry (2021), such bias terms are absorbed by the constant regressor.

²⁹ Our mechanism hinges on households' perceived joint distribution of aggregate variables, which may diverge from economic theory. As shown by Candia, Coibion, and Gorodnichenko (2020) and Han (2023), households typically associate higher inflation with lower real GDP growth. Our model reflects this by linking higher inflation with reduced real income, which broadly aligns with households' perceptions of inflation.

The CAAA preferences used above are nonhomothetic, and our mechanism relies on the interaction between nonhomotheticity and ambiguity to generate the disproportionate effect of inflation on poorer households (a notion similar to Cerreia-Vioglio, Maccheroni, and Marinacci 2022). Alternatively, we could assume that households have nonhomothetic preferences toward risk, due to borrowing limits or minimum consumption requirements. Then, homothetic preferences toward ambiguity would yield similar model predictions. Our assumptions are designed to align with our general theoretical framework, but this should be mainly seen as a convenient modeling device.

There are also reasonable alternative explanations for at least part of the evidence described above. For instance, Broer et al. (2022) show that the relationship between sensitivity to information and income levels can result from rational inattention, though this logic cannot explain the observed average bias. Another possibility is that households effectively report forecasts about the inflation rates of their own consumption baskets and that individual inflation rates differ by households' income levels. However, as shown in Kaplan and Schulhofer-Wohl (2017), the annual inflation difference between the top and bottom income groups is about 1 percent, which can only account for half of the difference in bias, while the persistence of inflation rates across households' income is virtually identical.³⁰ Yet another explanation is that individual forecasts are subject to exogenous shocks. This could help explain the observed pattern between forecast persistence and income levels, for instance, but only if these exogenous shocks are sufficiently correlated across individuals within income groups and are more volatile for poorer households. Overall, we think our theory and these alternatives are complementary, and they jointly depict a more complete picture of the expectation formation process. Our preference for the ambiguity aversion channel is due to its ability to simultaneously speak to the patterns of bias and persistence in a parsimonious framework.

IV. Application: Optimal Policy under Ambiguity

When agents in the private sector are ambiguity averse, how should policy respond to changes in fundamentals, and how does additional sensitivity and bias affect policy design? In this section, we explore a policy game that builds on Barro and Gordon (1983) to shed light on these questions.

A. Environment

Time-Consistent Policy Rule.—The policymaker chooses the inflation rate, π , so as to minimize the following social-loss function:

$$\mathcal{L} = E[U^2 + \omega(\pi - \pi^*)^2],$$

where U is the unemployment rate, π is the endogenous inflation rate, and π^* is an exogenous random inflation target that is drawn according to $\pi^* \sim \mathcal{N}(\bar{\pi}, \sigma_\pi^2)$.³¹

³⁰ See Table 3 in the main text and Figure 8 in the online Appendix of Kaplan and Schulhofer-Wohl (2017).

³¹ We normalize $\bar{\pi} = 0$ later on, but the main results do not hinge on this normalization.

The parameter ω balances the preference for lower unemployment against smaller deviations from the inflation target.

The policymaker faces a static Phillips curve that specifies how the unemployment rate reacts to average inflation surprises,

$$U = -\beta(\pi - \bar{\mathcal{F}}[\pi]),$$

where β is the slope of the Phillips curve. The unemployment rate is lower when actual inflation exceeds the average expected inflation. Importantly, agents in the private sector may not be rational, and the expectation operator corresponds to agents' subjective expectations. We have implicitly normalized the natural unemployment rate to zero, $U^* = 0$. Therefore, if $\pi = \bar{\mathcal{F}}[\pi] = \pi^*$, the policymaker would achieve the first-best outcome.

We consider a discretionary scenario in which the policymaker cannot commit to a policy *ex ante*. As in Barro and Gordon (1983), the time-consistent inflation policy is given by

$$(34) \quad \pi = (1 - \alpha)\pi^* + \alpha\bar{\mathcal{F}}[\pi], \text{ with } \alpha \equiv \frac{\beta^2}{\omega + \beta^2}.$$

That is, the inflation rate is a weighted average between the exogenous inflation target and the economy-wide subjective inflation expectation.

Subjective Expectations.—The exogenous inflation target cannot be perfectly observed by the public, and each agent i receives a private, noisy signal about it:

$$x_i = \pi^* + \epsilon_i, \text{ with } \epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2).$$

The noise here captures the notion that agents are inattentive to signals due to either attention costs or cognitive constraints. In addition to the informational frictions, agents in the private sector perceive ambiguity about the exogenous inflation target π^* . More specifically, agents believe that the mean of the target is ambiguous, that is,

$$\pi^* \sim \mathcal{N}(\mu, \sigma_\pi^2), \text{ and } \mu \sim \mathcal{N}(\bar{\pi}, \sigma_\mu^2),$$

where σ_μ controls the amount of ambiguity.

The payoff of an agent depends on their subjective expectation and on inflation itself,

$$u(\mathcal{F}_i[\pi], \pi) = -(\mathcal{F}_i[\pi] - \pi)^2 - \chi\pi.$$

That is, agents care about the accuracy of their forecast, and an increase in the inflation rate directly reduces their utility. This utility function can be seen as a reduced-form version of the microfounded structure introduced in Section III,³² which allows for biased subjective beliefs, in line with the data. In contrast with

³²For simplicity, we abstract from the complications due to the second-order term and effectively assume households are homogeneous except for the dispersed information.

Section III, inflation here is an endogenous equilibrium object that depends on the average subjective inflation expectation. As a result, the coordination motive is at play in shaping the subjective expectations.

B. Optimal Inflation Policy

In equilibrium, the optimal policy that satisfies condition (34) can be characterized by a pair of policy parameters, $\mathcal{R}$ and $\mathcal{C}$, such that

$$\pi = \mathcal{R}\pi^* + \mathcal{C},$$

where $\mathcal{R}$ represents the responsiveness to the inflation target and $\mathcal{C}$ determines the average level of inflation. How do aggregate inflation forecasts affect the optimal inflation policy?

We start with a rational expectations benchmark with informational frictions but in which agents do not perceive ambiguity.

PROPOSITION 10: *With rational expectations ($\sigma_\mu^2 = 0$), the optimal policy rule is given by*

$$\mathcal{R}^{RE} = 1 - \alpha + \alpha \frac{\sigma_\pi^2}{\sigma_\pi^2 + (1 - \alpha)^{-1} \sigma_\epsilon^2} \leq 1, \text{ and } \mathcal{C}^{RE} = 0.$$

Without ambiguity, the discretionary policy (34) can be viewed as a beauty contest game. When individual agents form expectations, they need to forecast the forecast of others, and the strength of these considerations is regulated by α . In the end, the average expectation is given by

$$\bar{E}[\pi] = \frac{\sigma_\pi^2}{\sigma_\pi^2 + (1 - \alpha)^{-1} \sigma_\epsilon^2} \pi^*,$$

where $(1 - \alpha)^{-1}$ captures the discounting of private signals due to the coordination motive. Effectively, agents form their expectations using (w, α) -modified signals as in Section II, with $w = 0$. Although the responsiveness is dampened due to dispersed information, the inflation rate remains proportional to the target, and $\mathcal{C}^{RE} = 0$. With full information rational expectations, the dampening effect vanishes, and the endogenous inflation perfectly tracks the target: $\mathcal{R}^{FIRE} = 1$.

Next, consider the case in which agents are ambiguity averse.

PROPOSITION 11: *With ambiguity aversion, the aggregate subjective expectation is given by*

$$\bar{\mathcal{F}}[\pi] = \mathcal{S}\pi^* + \mathcal{B},$$

where

$$\mathcal{S} = \frac{(1 + w)\sigma_\pi^2}{(1 + w)\sigma_\pi^2 + (1 - \alpha)^{-1}\sigma_\epsilon^2}, \quad \mathcal{B} = \chi\lambda(1 - \alpha + \alpha\mathcal{S})(1 - \mathcal{S})\sigma_\mu^2,$$

and

$$w = \frac{\sigma_\mu^2 / \sigma_\pi^2}{1 - 2\lambda(1 - \alpha)^2(1 - \mathcal{S})^2\sigma_\mu^2}.$$

The implied inflation policy satisfies

$$\mathcal{R} = 1 - \alpha + \alpha\mathcal{S} \in [\mathcal{R}^{RE}, 1], \text{ and } \mathcal{C} = \alpha\mathcal{B}.$$

Proposition 11 echoes our earlier emphasis on the effects of ambiguity aversion: the sensitivity to signals is amplified through the endogenously perceived prior, captured by the factor $(1 + w)$, and the expectation is permanently biased upward by $\mathcal{B}$.

The coordination motive, controlled by α , affects the equilibrium outcome in two ways. First, the policymaker internalizes the behavioral patterns of the private sector and adjusts the actual inflation process accordingly. This leads to higher responsiveness, $\mathcal{R} \geq \mathcal{R}^{RE}$, and a lifted intercept, $\mathcal{C} > 0$. The extent to which the inflation policy inherits this behavior from the private sector is mechanically increasing in the degree of coordination motive α . Second, the effect of ambiguity on an agent's subjective belief also hinges on the strength of the coordination motive, as both w and $\mathcal{B}$ depend on α directly. Different from the analysis in Section II, the economy here is inefficient even with complete information, and the socially optimal coordination level à la Angeletos and Pavan (2007) is $\hat{\alpha} \equiv 1 - (1 - \alpha)^2$, which is the key statistic that determines effects of ambiguity via w .³³

Figure 6 depicts the increased slope and shifted intercept that result from introducing ambiguity aversion. The red dashed line represents the inflation policy under FIRE, which aligns exactly with the 45-degree line. The black broken line displays the policy rule with noisy information but without ambiguity. Relative to FIRE, the only change is the dampened responsiveness. In both cases, there is no bias. When agents are ambiguity averse, we get the blue solid line. The slope of the policy approaches the one under FIRE, but this enhanced responsiveness is accompanied by a higher intercept due to the presence of bias in agents' forecasts.

C. Ambiguity and Social Welfare

Given an inflation policy rule, social welfare losses can be expressed as follows:

$$\mathcal{L} = E[U^2 + \omega(\pi - \pi^*)^2] = \frac{\omega}{\alpha}[(1 - \mathcal{R})^2\sigma_\pi^2 + \mathcal{C}^2].$$

The inflation policy under FIRE, with $\mathcal{R} = 1$ and $\mathcal{C} = 0$, achieves the first best. Further, either an increase in responsiveness, toward $\mathcal{R} = 1$, or a reduction in the magnitude of the intercept, $\mathcal{C}$, would improve welfare. Intuitively, the policymaker would like the public to pay attention to the inflation target, facilitating the implementation of the desired inflation level. Any irrelevant deviation of the average expectation from π^* is socially inefficient.

³³ We provide a detailed analysis of inefficient economies in online Appendix B.2.

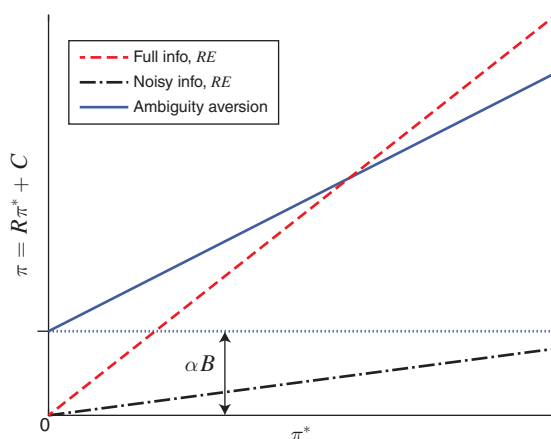


FIGURE 6. INFLATION POLICY WITH AND WITHOUT AMBIGUITY

When information is noisy, the responsiveness of the private sector is dampened, and the introduction of ambiguity can help approach the policy under FIRE. As can be observed in Figure 6, the combined effect of higher responsiveness and higher bias can lead to a better approximation of the FIRE first-best policy. This result is formalized in the following proposition.

PROPOSITION 12: *Fixing the amount of noise σ_ϵ^2 , there exists an intermediate level of ambiguity $\sigma_\mu^2 > 0$ that minimizes the social loss $\mathcal{L}$.*

Figure 7 helps illustrate the basic idea. Panel A shows how the responsiveness, $\mathcal{R}$, and the intercept, C , vary with the amount of ambiguity, σ_μ . On one hand, the introduction of ambiguity enhances the responsiveness to signals, as the private sector shifts attention from their ambiguous priors to their signals, which contain relevant information about the policy target. On the other hand, it generates bias, as the private sector becomes increasingly concerned about the realization of a high-inflation model. These two competing effects generate a *U*-shape social loss function, which is minimized at an intermediate level of ambiguity.

This mechanism could justify, for instance, the recent switch by the Fed from a fixed inflation target to average inflation targeting (AIT). Jia and Wu (2023) argue that the adoption of AIT by the Fed in 2020 introduced ambiguity about the target to the extent that the time horizon used to compute the average is unclear.³⁴

Alternative Type of Bounded Rationality.—So far, we have rationalized the bias in agents' forecasts with ambiguity aversion. However, other types of bounded rationality could lead to the observed bias, and this could lead to different policy implications.

³⁴Jia and Wu (2023) also show that the ambiguity generated by this switch can be beneficial, as it allows the Fed to increase its credibility. This benefit of ambiguity is different from the one we highlight. Their setup abstracts from ambiguity aversion, which plays a crucial role in our results.

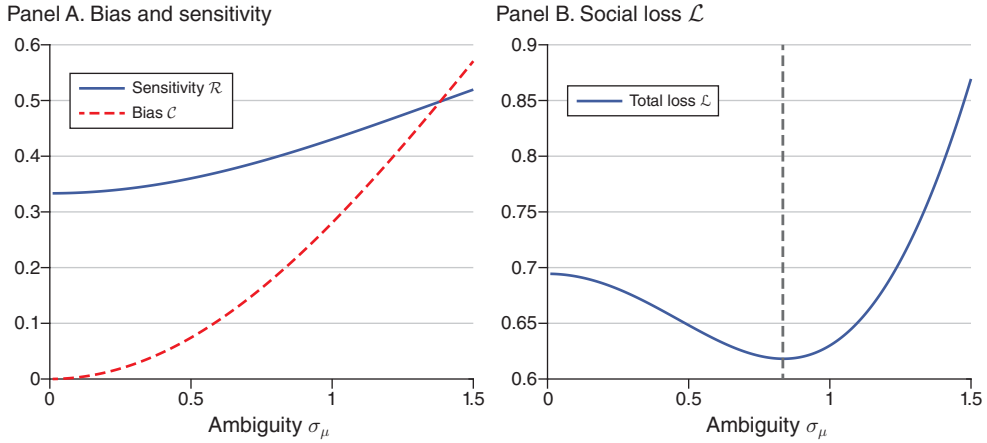


FIGURE 7. BIAS, SENSITIVITY, AND SOCIAL WELFARE

As an example, consider a heterogeneous prior approach à la Angeletos, Collard, and Dellas (2018). Specifically, suppose that each agent i understands that $\pi^* \sim \mathcal{N}(\bar{\pi}, \sigma_\pi^2)$ but believes that all other agents perceive the inflation target with a bias, $\pi^* \sim \mathcal{N}(\bar{\pi} + \mathcal{B}, \sigma_\pi^2)$. Further, suppose all agents continue to receive a noisy signal about the inflation target. In this case, without ambiguity, the optimal inflation policy also features a nonzero intercept, but the welfare implication is significantly different.

PROPOSITION 13: *With heterogeneous priors, the inflation policy rule is given by*

$$\mathcal{R} = \mathcal{R}^{RE}, \text{ and } \mathcal{C} = \alpha(\mathcal{R}^{RE} - 1)\mathcal{B}.$$

The social loss monotonically increases with $\mathcal{B}$.

In contrast to the results derived above, if the observed bias in inflation forecasts is attributed to heterogeneous priors, a higher bias necessarily implies lower welfare. Since there is no concomitant effect on the sensitivity to signals, bias simply leads to less accurate average expectations.

V. Connection with Robust Preference

In this section, we document an intimate connection between the smooth model of ambiguity and the robust preferences model (Hansen and Sargent 2001a,b). Although these two approaches to model uncertainty are conceptually different,³⁵ the main theoretical insights developed earlier about sensitivity and bias, along with the observational equivalence to Bayesian forecasts, also apply to models with robust preferences.

³⁵We refer to Hansen and Marinacci (2016) for a comprehensive discussion.

In parallel with the smooth model of ambiguity, we consider an efficient economy in which the utility function is given by

$$u(k_{it}, K_t, \xi_t) = -\frac{1}{2}[(1 - \alpha)(k_{it} - \xi_t)^2 + \alpha(k_{it} - K_t)^2] - \chi\xi_t - \frac{1}{2}\gamma\xi_t^2$$

and the signals follow the same general processes described in Section IIA.

We model robust preferences following Hansen and Sargent (2005). Agents worry about potential model misspecification and consider a set of alternatives:

$$(35) \quad \max_{k_{it}} \min_{m_{it}} E_{it} \left[u(k_{it}, K_t, \xi_t) m_{it} + \frac{1}{\varpi} m_{it} \log m_{it} \right]$$

subject to

$$m_{it} > 0, \text{ and } E_{it}[m_{it}] = 1.$$

Here, each random variable m_{it} introduces a distorted distribution, generating an alternative model, where $E_{it}[m_{it} \log m_{it}]$ corresponds to the relative entropy. The parameter ϖ controls the extent to which agents desire robustness. Agents then choose their strategies to optimize the worst-case scenario across the set of models under consideration.

Just as in the smooth model of ambiguity, under robust preferences, the subjective expectations and strategies of agents are jointly determined. Despite these complex interactions, the equilibrium strategy ultimately takes a simple form similar to the one in the smooth model.

PROPOSITION 14: *The linear strategy under robust preferences takes the following form:*

$$g(x_t^t) = (1 + r)p(L; w, \alpha)x_{it} + \mathcal{B}.$$

- (i) *The polynomial matrix $p(L; w, \alpha)$ is the Bayesian forecasting rule with the (w, α) -modified signal process, and w satisfies*

$$(36) \quad w = \frac{\varkappa_2}{1 - \alpha}.$$

- (ii) *The additional amplification, r , satisfies*

$$(37) \quad r = \frac{\varkappa_1 - \varkappa_2}{1 - \alpha + \varkappa_2}.$$

- (iii) *The level of bias, $\mathcal{B}$, satisfies*

$$(38) \quad \mathcal{B} = \chi \frac{r}{\bar{\gamma}}.$$

The two endogenous scalars $(\varkappa_1, \varkappa_2)$ are such that

$$\begin{aligned}\varkappa_1 - \varkappa_2 &= \varpi\gamma(1 - \alpha + \varkappa_1)V_{it}(\xi_t) + \varpi\gamma(\alpha - \varkappa_2)\text{cov}_{it}[K_t, \xi_t], \\ \varkappa_2 &= \frac{\alpha(1 - \alpha)}{\gamma}(\varkappa_1 - \varkappa_2) - \varpi\alpha(1 - \alpha)\text{disp}(k_{it}),\end{aligned}$$

where $V_{it}(\xi_t)$, $\text{cov}_{it}[K_t, \xi_t]$, and $\text{disp}(k_{it})$ denote the conditional volatility of the fundamental, the conditional covariance between the aggregate action and the fundamental, and the unconditional cross-sectional dispersion of individual actions, respectively.

Under robust preferences, agents want their actions to be robust across various models. In contrast, the smooth model of ambiguity is isomorphic to a setup where agents look for robustness across various priors about a fixed set of possible models (Hansen and Marinacci 2016). Despite the conceptual differences between these two approaches, Proposition 14 shows that they are, in a sense, observationally equivalent. The following corollary in turn provides the condition under which the two models are equivalent in terms of individual strategies, with only a difference in the amount of bias.

COROLLARY 3: *Fix the objective environment. For a robust preferences model that satisfies*

$$\begin{aligned}(i) \quad & w \geq 0, \quad r \geq 0, \quad \mathcal{S} \leq 1, \\ (ii) \quad & (1 - \mathcal{S}) \left[\frac{\gamma w}{(1 + w)r} - \frac{(1 - \alpha)(1 + w)r}{w} \right] + \gamma > (1 - \alpha) \frac{V(\xi_t - K_t)}{V(\xi_t)},\end{aligned}$$

there exists a smooth model of ambiguity such that the equilibrium strategies in these two models are identical up to a constant.

Although the equilibrium strategies share a similar transformation in terms of Bayesian expectations, there are still some subtle differences between the two approaches. In the smooth model of ambiguity, we focus on the case where the model uncertainty is only about the prior mean of the fundamental. With robust preferences, the model uncertainty is about the entire stochasticity of the environment. When determining the endogenous amplification parameter w , the former only requires unconditional moments of aggregate variables, while the latter requires conditional moments of both aggregate and individual variables.

Another difference between the two approaches is that the smooth model allows a separation between the attitude toward ambiguity, λ , and the amount of ambiguity, σ_μ , whereas the robust preferences model is more parsimonious, with a single parameter, ϖ , to control the overall concern about model misspecification. To demonstrate the quantitative difference between the two approaches, we revisit the inflation-expectations application. Specifically, we keep the objective environment fixed, including the exogenous stochastic process of inflation, ρ and σ_η , the private signal noise, σ_ϵ , and the preference parameters, β and ν . We examine the bias and persistence of forecast errors for households with median income. For the smooth

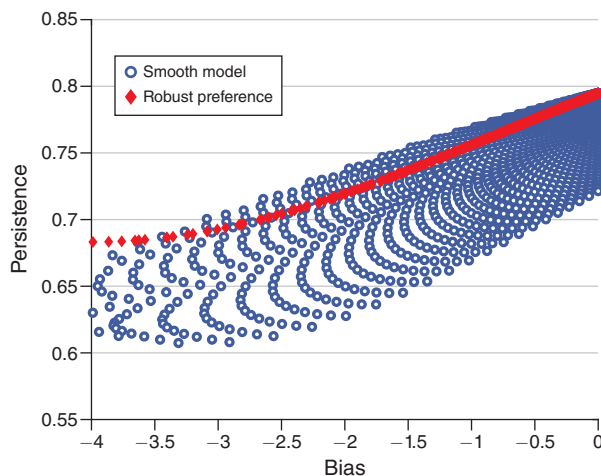


FIGURE 8. COMPARISON BETWEEN SMOOTH MODEL AND ROBUST PREFERENCES MODEL

Notes: This figure compares the predictions on the bias and persistence of forecast errors between the smooth model and the robust preferences model. Each blue circle corresponds to a smooth model with a choice of (λ, σ_μ^2) . Each red dot corresponds to a robust preferences model with a choice of ϖ . The objective environment is fixed the same as that in Section III.

model, we vary both the degree of ambiguity aversion, λ , and the amount of ambiguity, σ_μ^2 . For the robust preferences model, we vary the parameter controlling the preference for robustness, ϖ .

Figure 8 presents the joint distribution of bias and persistence of forecast errors. Each blue circle corresponds to the outcome under a specific pair of (λ, σ_μ^2) in the smooth model, while each red diamond point represents the outcome under a particular choice of ϖ in the robust preferences model. Both approaches yield the same qualitative prediction: higher bias is associated with lower persistence. Moreover, due to the additional degree of freedom in the smooth model, the scatterplot covers the line implied by the robust preferences model. Quantitatively, the two approaches also produce comparable predictions.

VI. Conclusion

In this paper, we study the effects of ambiguity in a general equilibrium environment with incomplete information. We provide an equivalence result that characterizes the equilibrium strategy as the solution to an adjusted single-agent Bayesian forecasting problem. Ambiguity aversion induces additional sensitivity to signals and a pessimistic bias, with both effects depending endogenously on agents' payoffs and the market structure. These properties allow us to match salient patterns observed in survey evidence on expectations, which are difficult to explain with rational expectation models. We also show that additional sensitivity and bias in subjective beliefs can change the effect of policy instruments and that the exact microfoundation for such belief distortions matters for policy design.

While our focus thus far has been on payoff structures that are homogeneous across agents, it would be interesting to explore the extent to which our main insights

can be extended to network games. For example, firms in production supply chains may have concerns about their upstream suppliers and downstream customers perceiving different models. Another direction for future research is to explore how policymakers should incorporate substantial deviations of subjective beliefs from rational ones into the design of monetary and fiscal policies in quantitative models.

REFERENCES

- Adam, Klaus.** 2007. "Optimal monetary policy with imperfect common knowledge." *Journal of Monetary Economics* 54 (2): 267–301.
- Altug, Sumru, Fabrice Collard, Cem Çakmaklı, Sujoy Mukerji, and Han Özşöylev.** 2020. "Ambiguous Business Cycles: A Quantitative Assessment." *Review of Economic Dynamics* 38: 220–37.
- Amador, Manuel, and Pierre-Olivier Weill.** 2010. "Learning from Prices: Public Communication and Welfare." *Journal of Political Economy* 118 (5): 866–907.
- Anderson, Evan W., Lars Peter Hansen, and Thomas J. Sargent.** 2003. "A Quartet of Semigroups for Model Specification, Robustness, Prices of Risk, and Model Detection." *Journal of the European Economic Association* 1 (1): 68–123.
- Andolfatto, David, Scott Hendry, and Kevin Moran.** 2008. "Are inflation expectations rational?" *Journal of Monetary Economics* 55 (2): 406–22.
- Angeletos, George-Marios, Fabrice Collard, and Harris Dellas.** 2018. "Quantifying Confidence." *Econometrica* 86 (5): 1689–1726.
- Angeletos, George-Marios, Zhen Huo, and Karthik Sastry.** 2021. "Imperfect Macroeconomic Expectations: Evidence and Theory." In *NBER Macroeconomics Annual*, Vol. 35, edited by Martin Eichenbaum and Erik Hurst, 1–86. Chicago, IL: University of Chicago Press.
- Angeletos, George-Marios, and Jennifer La'O.** 2010. "Noisy Business Cycles." In *NBER Macroeconomics Annual*, Vol. 24, edited by Daron Acemoglu, Kenneth Rogoff, and Michael Woodford, 319–78. Chicago, IL: University of Chicago Press.
- Angeletos, George-Marios, and Jennifer La'O.** 2020. "Optimal Monetary Policy with Informational Frictions." *Journal of Political Economy* 128 (3): 1027–64.
- Angeletos, George-Marios, and Alessandro Pavan.** 2007. "Efficient Use of Information and Social Value of Information." *Econometrica* 75 (4): 1103–42.
- Angeletos, George-Marios, and Karthik A. Sastry.** 2021. "Managing Expectations: Instruments versus Targets." *Quarterly Journal of Economics* 136 (4): 2467–2532.
- Azeredo da Silveira, Rava, Yeji Sung, and Michael Woodford.** 2024. "Optimally Imprecise Memory and Biased Forecasts." *American Economic Review* 114 (10): 3075–118.
- Backus, David, Axelle Ferriere, and Stanley Zin.** 2015. "Risk and Ambiguity in Models of Business Cycles." *Journal of Monetary Economics* 69: 42–63.
- Barro, Robert J., and David B. Gordon.** 1983. "Rules, Discretion and Reputation in a Model of Monetary Policy." *Journal of Monetary Economics* 12 (1): 101–21.
- Bhandari, Anmol, Jaroslav Borovicka, and Paul Ho.** 2024. "Survey Data and Subjective Beliefs in Business Cycle Models." *Review of Economic Studies* <https://doi.org/10.1093/restud/rdae054>.
- Bianchi, Francesco, Cosmin L. Ilut, and Martin Schneider.** 2017. "Uncertainty Shocks, Asset Supply and Pricing over the Business Cycle." *Review of Economic Studies* 85 (2): 810–54.
- Bianchi, Francesco, Sydney C. Ludvigson, and Sai Ma.** 2022. "Belief Distortions and Macroeconomic Fluctuations." *American Economic Review* 112 (7): 2269–2315.
- Bidder, R. M., and M. E. Smith.** 2012. "Robust Animal Spirits." *Journal of Monetary Economics* 59 (8): 738–50.
- Bordalo, Pedro, Nicola Gennaioli, Yueran Ma, and Andrei Shleifer.** 2020. "Overreaction in Macroeconomic Expectations." *American Economic Review* 110 (9): 2748–82.
- Broer, Tobias, and Alexandre N. Kohlhas.** 2024. "Forecaster (mis-) behavior." *Review of Economics and Statistics* 106 (5): 1334–51.
- Broer, Tobias, Alexandr Kohlhas, Kurt Mitman, and Kathrin Schlafmann.** 2022. "Expectation and Wealth Heterogeneity in the Macroeconomy." Unpublished.
- Candia, Bernardo, Olivier Coibion, and Yuriy Gorodnichenko.** 2020. "Communication and the Beliefs of Economic Agents." NBER Working Paper 27800.
- Cao, Shutao, Césaire A. Meh, José-Víctor Ríos-Rull, and Yaz Terajima.** 2021. "The Welfare Cost of Inflation Revisited: The Role of Financial Innovation and Household Heterogeneity." *Journal of Monetary Economics* 118: 366–80.

- Cerreia-Vioglio, Simone, Fabio Maccheroni, and Massimo Marinacci. 2022. "Ambiguity Aversion and Wealth Effects." *Journal of Economic Theory* 199: 104898.
- Cerreia-Vioglio, Simone, Roberto Corrao, and Giacomo Lanzani. 2024. "(Un-)Common Preferences, Ambiguity, and Coordination." Unpublished.
- Chen, Zengjing, and Larry Epstein. 2002. "Ambiguity, Risk, and Asset Returns in Continuous Time." *Econometrica* 70 (4): 1403–43.
- Coibion, Olivier, and Yuriy Gorodnichenko. 2015. "Information Rigidity and the Expectations Formation Process: A Simple Framework and New Facts." *American Economic Review* 105 (8): 2644–78.
- Collard, Fabrice, Sujoy Mukerji, Kevin Sheppard, and Jean-Marc Tallon. 2018. "Ambiguity and the Historical Equity Premium." *Quantitative Economics* 9 (2): 945–93.
- Das, Marcel, and Arthur van Soest. 1999. "A Panel Data Model for Subjective Information on Household Income Growth." *Journal of Economic Behavior and Organization* 40 (4): 409–26.
- Doepke, Matthias, and Martin Schneider. 2006. "Inflation and the Redistribution of Nominal Wealth." *Journal of Political Economy* 114 (6): 1069–97.
- Dominitz, Jeff. 1998. "Earnings Expectations, Revisions, and Realizations." *Review of Economics and Statistics* 80 (3): 374–88.
- Dominitz, Jeff, and Charles F. Manski. 1997. "Using Expectations Data to Study Subjective Income Expectations." *Journal of the American Statistical Association* 92 (439): 855–67.
- Elliott, Graham, Ivana Komunjer, and Allan Timmermann. 2008. "Biases in Macroeconomic Forecasts: Irrationality or Asymmetric Loss?" *Journal of the European Economic Association* 6 (1): 122–57.
- Epstein, Larry G., and Tan Wang. 1994. "Intertemporal Asset Pricing under Knightian Uncertainty." *Econometrica* 62 (2): 283–322.
- Farmer, Leland, Emi Nakamura, and Jon Steinsson. 2024. "Learning About the Long Run." *Journal of Political Economy* (10): 3334–77.
- Gallant, A. Ronald, Mohammad R. Jahan-Parvar, and Hening Liu. 2018. "Does Smooth Ambiguity Matter for Asset Pricing?" *Review of Financial Studies* 32 (9): 3617–66.
- Gilboa, Itzhak, and David Schmeidler. 1989. "Maxmin Expected Utility with Non-Unique Prior." *Journal of Mathematical Economics* 18 (2): 141–53.
- Hanany, Eran, and Peter Klibanoff. 2009. "Updating Ambiguity Averse Preferences." *B.E. Journal of Theoretical Economics* 9 (1): <https://doi.org/10.2202/1935-1704.1547>.
- Hanany, Eran, Peter Klibanoff, and Sujoy Mukerji. 2020. "Incomplete Information Games with Ambiguity Averse Players." *American Economic Journal: Microeconomics* 12 (2): 135–87.
- Hansen, Lars Peter, and Massimo Marinacci. 2016. "Ambiguity Aversion and Model Misspecification: An Economic Perspective." *Statistical Science* 31 (4): 511–15.
- Hansen, Lars Peter, and Thomas J. Sargent. 2001a. "Acknowledging Misspecification in Macroeconomic Theory." *Review of Economic Dynamics* 4 (3): 519–35.
- Hansen, Lars Peter, and Thomas J. Sargent. 2001b. "Robust Control and Model Uncertainty." *American Economic Review* 91 (2): 60–66.
- Hansen, Lars Peter, and Thomas J. Sargent. 2005. "Robust Estimation and Control Under Commitment." *Journal of Economic Theory* 124 (2): 258–301.
- Hansen, Lars Peter, Thomas J. Sargent, and Thomas D. Tallarini. 1999. "Robust Permanent Income and Pricing." *Review of Economic Studies* 66 (4): 873–907.
- Han, Zhao. 2023. "Asymmetric Information and Misaligned Inflation Expectations." *Journal of Monetary Economics* 143: 103529.
- Huo, Zhen, and Marcelo Zouain Pedroni. 2020. "A Single-Judge Solution to Beauty Contests." *American Economic Review* 110 (2): 526–68.
- Huo, Zhen, Marcelo Pedroni, and Guangyu Pei. 2024. *Data and Code for: "Bias and Sensitivity under Ambiguity."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E208166V1>.
- Ilut, Cosmin, and Hikaru Saijo. 2021. "Learning, Confidence, and Business Cycles." *Journal of Monetary Economics* 117: 354–76.
- Ilut, Cosmin, and Martin Schneider. 2014. "Ambiguous Business Cycles." *American Economic Review* 104 (8): 2368–99.
- Ilut, Cosmin, and Martin Schneider. 2023. "Modeling Uncertainty as Ambiguity: A Review." In *Handbook of Economic Expectations*, edited by Rüdiger Bachmann, Giorgio Topa, and Wilbert van der Klaauw, 749–77. London: Academic Press.
- Jia, Chengcheng, and Jing Cynthia Wu. 2023. "Average Inflation Targeting: Time Inconsistency and Ambiguous Communication." *Journal of Monetary Economics* 138: 69–86.
- Ju, Nengjiu, and Jianjun Miao. 2012. "Ambiguity, Learning, and Asset Returns." *Econometrica* 80 (2): 559–91.

- Kaplan, Greg, and Sam Schulhofer-Wohl.** 2017. "Inflation at the Household Level." *Journal of Monetary Economics* 91: 19–38.
- Klibanoff, Peter, Massimo Marinacci, and Sujoy Mukerji.** 2005. "A Smooth Model of Decision Making Under Ambiguity." *Econometrica* 73 (6): 1849–92.
- Kohlhas, Alexandre, and Ansgar Walther.** 2021. "Asymmetric Attention." *American Economic Review* 111 (9): 2879–2925.
- Kohlhas, Alexandre, and Donald Robertson.** 2024. "Cautious Expectations." Unpublished.
- Lorenzoni, Guido.** 2010. "Optimal Monetary Policy with Uncertain Fundamentals and Dispersed Information." *Review of Economic Studies* 77 (1): 305–38.
- Luo, Yulei, and Eric R. Young.** 2010. "Risk-Sensitive Consumption and Savings under Rational Inattention." *American Economic Journal: Macroeconomics* 2 (4): 281–325.
- Marinacci, Massimo.** 2015. "Model Uncertainty." *Journal of the European Economic Association* 13 (6): 1022–1100.
- Massenot, Baptiste, and Yuri Pettinicchi.** 2019. "Can Households See into the Future? Survey Evidence from the Netherlands." *Journal of Economic Behavior and Organization* 164: 77–90.
- Miao, Jianjun.** 2009. "Ambiguity, Risk and Portfolio Choice under Incomplete Information." *Annals of Economics and Finance* 10 (2): 257–79.
- Michelacci, Claudio, and Luigi Paciello.** 2019. "Ambiguous Policy Announcements." *Review of Economic Studies* 87 (5): 2356–98.
- Michelacci, Claudio, and Luigi Paciello.** 2024. "Ambiguity Aversion and Heterogeneity in Households' Beliefs." *American Economic Journal: Macroeconomics* 16 (2): 95–126.
- Molavi, Pooya.** 2023. "Simple Models and Biased Forecasts." Unpublished.
- Morris, Stephen, and Hyun Song Shin.** 2002. "Social Value of Public Information." *American Economic Review* 92 (5): 1521–34.
- Paciello, Luigi, and Mirko Wiederholt.** 2014. "Exogenous Information, Endogenous Information, and Optimal Monetary Policy." *Review of Economic Studies* 81 (1): 356–88.
- Pei, Guangyu.** 2024. "Pessimism, Disagreement, and Economic Fluctuations." *Journal of the European Economic Association* 22 (3): 1177–1227.
- Pope, Devin G., and Maurice E. Schweitzer.** 2011. "Is Tiger Woods Loss Averse? Persistent Bias in the Face of Experience, Competition, and High Stakes." *American Economic Review* 101 (1): 129–57.
- Rozsypal, Filip, and Kathrin Schlafmann.** 2023. "Overpersistence Bias in Individual Income Expectations and Its Aggregate Implications." *American Economic Journal: Macroeconomics* 15 (4): 331–71.
- Sung, Yeji.** 2024. "Macroeconomic Expectations and Cognitive Noise." Unpublished.
- Townsend, Robert M.** 1983. "Forecasting the Forecasts of Others." *Journal of Political Economy* 91 (4): 546–88.
- University of Michigan, Survey Research Center, Surveys of Consumers.** 1987–2020. "Micro data: Michigan Survey of Consumers." <https://sda.umsurvey.org/sda-public/cgi-bin/hsda2?setupfile=harcsda&datasetname=sca&ui=2&action=subset> (accessed July 12, 2024).
- US CPI Inflation.** 1987–2021. "Consumer Price Index for All Urban Consumers: All Items in U.S. City Average." Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL> (accessed July 12, 2024).
- Woodford, Michael.** 2003. "Imperfect Common Knowledge and the Effects of Monetary Policy." In *Knowledge, Information, and Expectations in Modern Macroeconomics*, edited by Philippe Aghion, Roman Frydman, Joseph Stiglitz, and Michael Woodford, 25–58. Princeton, NJ: Princeton University Press.